A COMPANION TO THE PROOFS

Mathematical Background

for MIACE and Monotone Tracking

Measure theory, functional analysis, Laplace transforms,
complex analysis, interpolation, nonlinear dynamics and delay equations

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These notes accompany *MIACE and Monotone Tracking: Exact Tuning for First-Order Plants and Positive Cascades with Pure Delay.* They explain the analytical tools behind its proofs, develop worked examples, and provide a reading map for all 32 numbered results, including the nonlinear extension of 22 October 2026.

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Contents

1 How to read the proofs **1**

1.1 The manuscript and the scope of the companion 1
1.2 A route through the material 2
1.3 Notation and the objects being integrated 2
1.4 Four distinctions to keep visible 3
1.5 Background references 3

2 Integration, signs, and accumulated motion **4**

2.1 What an integral ignores, and what it detects 4
2.2 The signal spaces and their elementary inequalities 5
2.3 Passing to limits without losing an area 5
2.4 Interchanging integrations 6
2.5 Absolute continuity connects derivatives to motion 7
2.6 Total variation and the cost of reversing direction 8
2.7 The strict inequality at the attainment boundary 9
2.8 Two exercises with solutions 9

3 Convolution, positivity, and stable inversion **11**

3.1 Why completeness is needed 11
3.2 Convolution as an operator 12
3.3 Finite measures include feedthrough and exact delay 13
3.4 The geometric series becomes an inverse 14
3.5 Exponential weights and actual decay 15
3.6 Why a small delay becomes small after a lag 16
3.7 From signal perturbations to MIACE perturbations 17
3.8 Two exercises with solutions 18

4 Laplace transforms, analytic continuation, and exact delay remainders **19**

4.1 Existence and the elementary transform rules 19
4.2 Exponential moments and holomorphic transforms 20
4.3 The small part of complex analysis that is needed 21
4.4 Hermite interpolation and the simplex formula 22
4.5 An exact compactly supported remainder 23
4.6 Exercises with solutions 24

5 Differential equations, delay, and quadratic energy **26**

5.1 Linear ODEs with ordinary inputs 26
5.2 Why the equality signs imply monotonicity in a cascade 27
5.3 Exponential stability and the role of the realization 27

5.4 Retarded delay equations and the method of steps 28
5.5 Stable internal-model realizations 30
5.6 Quadratic energy as an inner-product calculation 30
5.7 Gramians and the second-order energy formula 31
5.8 A delayed energy calculation through autocorrelation 32
5.9 Exercises with solutions 32

6 From analytical tools to optimal controllers 34
6.1 A balance law and a weighted distance 34
6.2 Efficiency, attainability, and uniqueness 35
6.3 The earliest admissible pulse minimizes an increasing moment 36
6.4 Signs of two exponentials and repeated poles 36
6.5 Taylor expansion of a squared residual 38
6.6 Exercises with solutions 38

7 Nonlinear storage balances and exact MIACE tuning 40
7.1 A chain rule that can be integrated 40
7.2 Local solutions and why a bounded trajectory continues 41
7.3 The nonlinear balance and the unchanged area geometry 41
7.4 Constructing an attaining feedback 42
7.5 A nonlinear example with an exact implicit trajectory 43
7.6 Finite-horizon integration and its omitted tail 43
7.7 Exercises with solutions 44

8 A reading map for every numbered result 45
8.1 The 32 results 45
8.2 The appendices and the numerical section 48
8.3 What is proved here, and what remains standard background 49

References and further reading 50

Chapter 1

How to read the proofs

A unit step asks a system to move from one equilibrium to another. The error records the distance still to be covered. The centered command records the extra command supplied during the move. Integrating their absolute values gives two areas, and MIACE measures their weighted distance from the origin. That picture is elementary. The questions behind it become more demanding: when is the distance minimal, what does equality force the signals to do, and can a stable controller produce those signals while retaining a pure delay?

The companion follows those questions. The first proofs need a precise version of familiar integration rules. The later constructions need convergent series of convolution operators. Complex analysis explains why equality forces cancellations, and interpolation supplies an exact representation of the delay. Linear systems theory connects these representations back to internal stability.

1.1 The manuscript and the scope of the companion

The reference manuscript is the author's edited v2 with the nonlinear extension integrated on 22 October 2026. Its numbered statements run from Theorem 1 to Theorem 32, with lemmas and propositions sharing the same sequence. Results 1–31 retain their earlier numbers; Theorem 32 and paper Section 15 develop a nonlinear first-order class without delay. Throughout these notes, "manuscript Theorem 17" refers to that paper; "Theorem 3.4", for example, refers to this companion. The complete map is in chapter 8. The companion does not change the manuscript's plant classes, controller classes, or conclusions.

Most elementary tools used here are proved. A few foundational results are stated as standard inputs with references: the construction of Lebesgue measure, the fundamental characterization of absolute continuity, uniqueness of the Laplace transform, and the spectral and oscillation theorems for retarded differential equations. Their hypotheses and the points where they enter the argument are made explicit. Proving all of their foundations would require separate courses in real analysis and functional differential equations.

The expected starting point is calculus of one and several variables, linear algebra, complex numbers, and elementary differential equations. For a first reading, it is enough to work on the half-line with Lebesgue measure. General measurable spaces enter only where they make a standard integration rule clearer. Hahn–Banach, general spectral decompositions, weak compactness and a theory of distributions are outside the required route.

1.2 A route through the material

Question in the proof	Background to learn	Manuscript location
What do absolute areas measure?	Positive and negative parts; equality in the integral triangle inequality.	Sections 3–5 and 7.1
Why is the output monotone?	Absolute continuity; the fundamental theorem of calculus; positivity of lag kernels.	Sections 4, 5, 7.2 and 8.1
Why does a constructed response exist?	Completeness of L^1; Young's inequality; the convolution Neumann series.	Sections 6.2 and 7.6
Why does it decay?	Exponential weights, weighted supremum estimates, and the stability theorem for retarded realizations.	Sections 6–8
Why must a pole cancel?	Laplace analyticity, positivity on the real axis, and the identity theorem.	Sections 5.4 and 6.1
How is the exact delay retained?	Hermite interpolation, repeated poles and finite-support kernels.	Sections 7.5–7.8
What changes under a cap or a perturbation?	Increasing-weight rearrangement; finite signed measures; norm estimates.	Sections 8.3–8.4 and 12–13
How does the argument extend to a nonlinear plant?	The chain rule for absolutely continuous functions; scalar ODE existence, continuation and exponential bounds; storage balance.	Section 15

For the first pass, read Chapters 2 and 3, then return to manuscript Lemma 9. Its proof is short, but it uses the essential chain: an integral inequality bounds every term, a geometric series makes the terms summable, completeness gives a limit, and continuity of the convolution operator lets the limit solve the equation. The positive signs survive throughout that process.

For the second pass, read Chapters 4 and 5. Chapter 6 then connects these tools to optimization, caps, pole signs and mismatch. The exercises include solutions so that the notes can be used independently of a course.

For manuscript Theorem 32, continue to Chapter 7. It derives the storage balance for $y' = a(y)(w - y)$, explains why the same absolute-area argument applies, and verifies that the proposed feedback produces a global admissible response. The original error $1 - y$ and the fixed time variable are retained. No nonlinear transport delay is included in that result.

1.3 Notation and the objects being integrated

We use t for the time variable in general analysis statements. In a normalized manuscript calculation it represents $\theta = t_{\mathrm{phys}}/\tau$ or t_{phys}/τ_*, as specified there. The Laplace variable is p. The distinction between physical and normalized time must be retained when numerical gains or dimensional areas are recovered. The nonlinear section uses a fixed reference time scale. Its storage coordinate does not introduce a state-dependent change of time.

Notation	Meaning
$e = 1 - y$	Error for a positive normalized unit step.
$w = ku$, $v = w - 1$	Plant-scaled command and its deviation from the new equilibrium.
A_e, A_u	$\int_0^\infty \lvert e\rvert$ and $\int_0^\infty \lvert v\rvert$.
$\mathcal{M}_\alpha$	$\sqrt{A_e^2 + \alpha A_u^2}$, with $\alpha > 0$.
f_+, f_-	$\max(f, 0)$ and $\max(-f, 0)$.
$\widehat{f}(p)$	$\int_0^\infty e^{-pt} f(t)\,\mathrm{d}t$, where it exists.
$K * f$	Causal convolution, $\int_0^t K(t-s)f(s)\,\mathrm{d}s$.
δ_T	Unit point mass at time T; δ_0 is the convolution identity.
$\lVert\nu\rVert_{\mathrm{TV}}$	Total variation of a finite signed impulse measure.
δ	The manuscript's normalized transport delay; it is distinct from δ_T.
$\Phi_a(y)$, S_a	The nonlinear storage coordinate $\int_0^y a(s)^{-1}\,\mathrm{d}s$ and its equilibrium change $\Phi_a(1)$.

1.4 Four distinctions to keep visible

Signed area and absolute area. The transform at zero gives $\widehat{f}(0) = \int f$ for $f \in L^1$. It gives $\int |f|$ only after nonnegativity has been established. Replacing one by the other before proving the sign would remove the very penalty that MIACE is intended to measure.

An almost-everywhere assertion and a trajectory assertion. Integral equality identifies an L^1 equivalence class. An output trajectory has a particular continuous representative supplied by the differential equation. This regularity permits the passage from almost- everywhere signs to statements about every time.

A signal and a filter's impulse response. The physical commands in the paper are ordinary functions, including functions with finite jumps. A proper filter can nevertheless have a Dirac mass in its impulse response: convolution of that mass with an ordinary input is simply direct feedthrough. An impulse response and the response to a step are different objects.

A cost minimum and a transient shape. The two areas do not determine the whole trajectory. At $\alpha = 1$, the two admissible first-order pairs

$$e_1 = e^{-2t}, \quad v_1 = e^{-2t}, \qquad e_2 = \tfrac{1}{4}e^{-t} + \tfrac{3}{4}e^{-3t}, \quad v_2 = \tfrac{3}{2}e^{-3t}$$

both satisfy $e' = -e - v$, and both have $A_e = A_u = 1/2$. Their tails decay at different rates. The paper's uniqueness assertion for the positive PI family therefore contains information about that controller class in addition to the two integral values.

1.5 Background references

The basic integration and normed-space material can be studied alongside the free MIT notes [11, 12]. The transform and complex-analysis material is covered by Orloff [10]. For the interpolation used in the exact delay remainder, the relevant classical reference is de Boor [3], especially the integral representations. Convolution resolvents and functional differential equations are developed more broadly in Gripenberg et al. [7] and Hale and Verduyn Lunel [8]. These references support the background tools; the theorem map identifies how the manuscript uses them.

Chapter 2

Integration, signs, and accumulated motion

A command may jump, a derivative may fail to exist at isolated times, and an integral may extend over an infinite interval. None of these facts is unusual in the manuscript. The purpose of Lebesgue integration here is to keep such statements precise while allowing the usual calculations with areas. This chapter develops the facts used in manuscript Theorems 2–5, 12, 15, and 20. Throughout, time belongs to $\mathbb{R}_+ = [0, \infty)$ and integration without a specified measure means Lebesgue integration.

2.1 What an integral ignores, and what it detects

A *σ-algebra* is a collection of subsets containing the whole space and closed under complements and countable unions. Its members are the measurable sets. A positive measure assigns a number in $[0, \infty]$ to each measurable set, assigns zero to the empty set, and is countably additive on disjoint sets. Lebesgue measure, denoted by m, assigns an interval its length. Completeness means that every subset of a null set is measurable. Its existence, completeness, and the usual translation and change-of-scale properties are background facts of measure theory; their construction is not repeated here [12].

A real function f is measurable when $\{t : f(t) > a\}$ is measurable for every real a. Continuous and piecewise continuous signals are measurable. Limits, sums, products, absolute values, and positive parts of measurable functions are measurable wherever the expressions are defined. A property holds *almost everywhere*, abbreviated a.e., when the set on which it fails has measure zero. A singleton and every countable set have measure zero.

For a nonnegative simple function $s = \sum_{j=1}^{n} a_j \mathbf{1}_{E_j}$, with disjoint measurable E_j and $a_j \geq 0$, define

$$\int s = \sum_{j=1}^{n} a_j m(E_j).$$

For a nonnegative measurable f, define $\int f$ as the supremum of $\int s$ over all nonnegative simple $s \leq f$. The value may be infinite. For a real function, put

$$f_+ = \max\{f, 0\}, \qquad f_- = \max\{-f, 0\}.$$

If $\int |f| < \infty$, both parts have finite integral and $\int f = \int f_+ - \int f_-$. This is absolute integrability.

Lemma 2.1 (Vanishing of a nonnegative integral)**.** *If $g \geq 0$ is measurable, then $\int g = 0$ if and only if $g = 0$ a.e.*

Proof. For every positive integer n, let $E_n = \{g \geq 1/n\}$. Monotonicity of the integral gives $m(E_n)/n \leq \int g$. If the latter is zero, every E_n is null. Their union is $\{g > 0\}$ and is null by countable subadditivity. Conversely, an integral over a null set is zero, directly from the definition through simple functions. □

Proposition 2.2 (Absolute area and negative area)**.** *For a real integrable f,*

$$\int |f| = \int f + 2 \int f_-. \tag{2.1}$$

In particular, $\int |f| = \int f$ if and only if $f \geq 0$ a.e.

Proof. The pointwise identities $f = f_+ - f_-$ and $|f| = f_+ + f_-$ give $|f| = f + 2f_-$. Integrate and apply Lemma 2.1 to f_-. □

This is the source of the negative-area penalty in the manuscript. For the error, $e_- = \max\{y - 1, 0\}$ records output overshoot. For the centered command, v_- records the command's excursion below its new equilibrium. Equality of an absolute area with its signed area therefore supplies a sign condition, not merely a numerical coincidence. If a continuous function is nonnegative a.e., it is nonnegative everywhere: a negative value would, by continuity, give a whole interval of negative values. Right continuity gives the same conclusion at a finite command jump when the right-hand value is the chosen convention.

Example 2.3 (Cancellation can hide an excursion)**.** Let $f = 2\mathbf{1}_{[0,1)} - \mathbf{1}_{[1,2)}$. Its signed integral is 1, its absolute integral is 3, and its negative-part integral is 1. Thus $3 = 1 + 2 \cdot 1$. Changing f at $t = 1$ changes none of these integrals. A finite jump and an impulse are different: the jump leaves an ordinary function, whereas a Dirac point mass belongs to a space of measures discussed later.

2.2 The signal spaces and their elementary inequalities

For $1 \le p < \infty$, the space $L^p(\mathbb{R}_+)$ consists of measurable functions with $\int |f|^p < \infty$, with functions agreeing a.e. identified. Its norm is

$$\|f\|_p = \left(\int_0^\infty |f(t)|^p \, \mathrm{d}t \right)^{1/p}.$$

The norm on L^∞ is the *essential supremum*: the smallest $M \ge 0$ such that $|f| \le M$ a.e. Identifying functions a.e. makes $\|f\|_p = 0$ equivalent to $f = 0$ as an element of the space. Values at an initial time are specified separately by continuity, right limits, or initial conditions.

The inequalities needed for absolute areas follow from pointwise inequalities:

$$\left| \int f \right| \le \|f\|_1, \qquad \|f + g\|_1 \le \|f\|_1 + \|g\|_1, \qquad \big| \|f\|_1 - \|g\|_1 \big| \le \|f - g\|_1. \tag{2.2}$$

For real f, the first follows by integrating $-|f| \le f \le |f|$. The second follows from $|f+g| \le |f|+|g|$. Apply the second to $f = (f - g) + g$, then interchange f and g, to obtain the third. The corresponding complex inequality follows by multiplying $\int f$ by a unit complex number that makes it nonnegative real and then taking real parts inside the integral.

On an infinite interval, L^1 and L^2 have no general inclusion in either direction. For example, $t^{-2/3}\mathbf{1}_{(0,1)}$ belongs to L^1 but not L^2, whereas $(1 + t)^{-3/4}$ belongs to L^2 but not L^1. Additional boundedness does give the useful implication

$$f \in L^1 \cap L^\infty \quad \Longrightarrow \quad \|f\|_2^2 \le \|f\|_\infty \|f\|_1.$$

The manuscript states integrability assumptions explicitly because an infinite-horizon area and an infinite-horizon squared area are different requirements.

2.3 Passing to limits without losing an area

Theorem 2.4 (Monotone convergence)**.** *If $0 \le f_n \uparrow f$ a.e., then $\int f_n \uparrow \int f$, allowing the value $+\infty$.*

Proof. Changing the functions on a null set lets us suppose the inequalities hold everywhere. Monotonicity gives a limit $L = \lim_n \int f_n \le \int f$. Fix a nonnegative simple function $s \le f$ and $0 < c < 1$. On the set where $s > 0$, the increasing sets $E_n = \{f_n \ge cs\}$ exhaust that set. Thus $\int f_n \ge c \int s\mathbf{1}_{E_n}$, and countable additivity of measure, applied to the finitely many levels of s, gives $L \ge c \int s$. Let $c \uparrow 1$ and take the supremum over s. This yields $L \ge \int f$. □

Lemma 2.5 (Fatou's lemma)**.** *For nonnegative measurable functions f_n,*

$$\int \liminf_{n\to\infty} f_n \le \liminf_{n\to\infty} \int f_n.$$

Proof. Put $h_n = \inf_{k\ge n} f_k$. Then h_n is measurable, $h_n \uparrow \liminf f_n$, and $\int h_n \le \inf_{k\ge n} \int f_k$. Monotone convergence gives the result on taking limits. □

Theorem 2.6 (Dominated convergence)**.** *Suppose $f_n \to f$ a.e. and $|f_n| \le g$ a.e. for a single nonnegative $g \in L^1$. Then $f \in L^1$, $\|f_n - f\|_1 \to 0$, and $\int f_n \to \int f$.*

Proof. The pointwise limit gives $|f| \le g$, so $f \in L^1$. Apply Fatou's lemma to the nonnegative functions $2g - |f_n - f|$. Their pointwise limit is $2g$, and hence

$$2\int g \le 2\int g - \limsup_n \int |f_n - f|.$$

The integrals of g are finite, so this implies $\limsup_n \|f_n - f\|_1 \le 0$. The last conclusion follows from $|\int (f_n - f)| \le \|f_n - f\|_1$. □

For $f \in L^1$, dominated convergence applied to $\mathbf{1}_{[0,T]} f$ gives

$$\lim_{T\to\infty} \int_0^T f = \int_0^\infty f, \qquad \lim_{T\to\infty} \int_T^\infty |f| = 0. \tag{2.3}$$

It also justifies evaluation of a Laplace transform at zero from the right: $e^{-pt} f(t) \to f(t)$ as $p \downarrow 0$, with absolute value bounded by $|f|$. The same argument is used in manuscript Theorem 12.

Example 2.7 (Why the bound must be integrable)**.** The functions $f_n = \mathbf{1}_{[n,n+1]}$ converge pointwise to zero, but every integral equals one. An integrable common bound cannot exist. Fatou remains valid and gives $0 \le 1$; equality of the limiting integrals would be false. This is the elementary reason to check the hypotheses each time a limit crosses an integral sign.

2.4 Interchanging integrations

Theorem 2.8 (Tonelli and Fubini on the time half-plane)**.** *Let F be measurable on $\mathbb{R}^2_+$. If $F \ge 0$, then*

$$\int_{\mathbb{R}^2_+} F(s,t)\,\mathrm{d}(s,t) = \int_0^\infty \left(\int_0^\infty F(s,t)\,\mathrm{d}s \right) \mathrm{d}t = \int_0^\infty \left(\int_0^\infty F(s,t)\,\mathrm{d}t \right) \mathrm{d}s,$$

with infinite values permitted. If $\int_{\mathbb{R}^2_+} |F| < \infty$, the same identities hold for real or complex F with finite integrals, and the sections are integrable for almost every value of the other variable.

This is a background theorem of product integration [12]. Its construction starts with indicators of measurable rectangles, extends to nonnegative measurable functions by approximation and monotone convergence, and then applies the nonnegative result to the positive and negative parts. The absolute-integrability hypothesis makes both resulting integrals finite, so their difference is legitimate. In the applications below, one can see the required sign or absolute bound directly.

Example 2.9 (A convolution mass and a tail moment)**.** Let $K, f \ge 0$ be integrable causal functions. On the triangle $0 \le s \le t$, Tonelli and the substitution $u = t - s$ give

$$\begin{aligned} \int_0^\infty (K * f)(t)\,\mathrm{d}t &= \int_0^\infty \int_s^\infty K(t-s) f(s)\,\mathrm{d}t\,\mathrm{d}s \\ &= \left(\int_0^\infty K(u)\,\mathrm{d}u \right) \left(\int_0^\infty f(s)\,\mathrm{d}s \right). \end{aligned}$$

For an integrable $J \geq 0$ supported in $[0, \delta]$, define $K(t) = \int_t^\delta J(r)\,\mathrm{d}r$ on that interval and zero afterwards. The same interchange yields

$$\int_0^\delta K(t)\,\mathrm{d}t = \int_0^\delta \int_0^r J(r)\,\mathrm{d}t\,\mathrm{d}r = \int_0^\delta rJ(r)\,\mathrm{d}r.$$

Thus the mass of the tail kernel is the first moment of J. This is the relation $\int K = j_1$ in the manuscript's finite-support delay construction. No differentiation of J is required.

2.5 Absolute continuity connects derivatives to motion

Definition 2.10. A function $f : [a, b] \to \mathbb{R}$ is absolutely continuous if, for every $\varepsilon > 0$, there is an $\eta > 0$ such that every finite disjoint collection of subintervals (a_j, b_j) with $\sum_j (b_j - a_j) < \eta$ satisfies $\sum_j |f(b_j) - f(a_j)| < \varepsilon$. It is locally absolutely continuous on $\mathbb{R}_+$ if it has this property on every bounded interval.

Theorem 2.11 (Fundamental theorem for absolutely continuous functions)**.** *A function f on $[a, b]$ is absolutely continuous if and only if there is a $g \in L^1(a, b)$ such that*

$$f(t) = f(a) + \int_a^t g(s)\,\mathrm{d}s.$$

In that case f' exists a.e., $f' = g$ a.e., and $f(t) - f(s) = \int_s^t f'(r)\,\mathrm{d}r$ for all $a \leq s \leq t \leq b$.

This fundamental theorem uses the Lebesgue differentiation theorem and the characterization of absolute continuity. We take those foundational results from real analysis [5, 12]. The direction most visible in an ODE is worth explaining. For $g \in L^1$, the integral over a measurable set of sufficiently small measure is uniformly small: truncate $|g|$ at a large level and then control the bounded part by the measure of the set. Applying this to the union of disjoint intervals proves absolute continuity of its indefinite integral. Lebesgue differentiation identifies its derivative with g a.e.

Corollary 2.12 (Derivative signs and endpoint identities)**.** *If f is locally absolutely continuous and $f' \geq 0$ a.e., then f is nondecreasing. For any locally absolutely continuous f, regardless of the sign of its derivative, $f' \in L^1(\mathbb{R}_+)$ implies that f has a finite limit at infinity and*

$$f(\infty) - f(0) = \int_0^\infty f'(t)\,\mathrm{d}t.$$

If also $f \in L^1$, that limit is zero.

Proof. For $b > a$, the fundamental theorem gives $f(b) - f(a) = \int_a^b f' \geq 0$. Without a sign assumption, it gives $|f(b) - f(a)| \leq \int_a^b |f'|$. The tails of an L^1 function tend to zero, so the latter bound makes $f(t)$ Cauchy as $t \to \infty$. Pass to the limit in the endpoint identity. A nonzero limiting value would keep $|f|$ bounded below by a positive constant on an infinite tail, contradicting $f \in L^1$. □

This corollary explains the role of the plant equation. If $e' = -e - v$ and $e, v \in L^1$, then $e' \in L^1$ as well. With local absolute continuity, the endpoint limit can therefore be deduced. In settings where these hypotheses have not been established, the endpoint condition must be supplied or proved separately.

On bounded intervals, products of absolutely continuous functions are absolutely continuous and satisfy the product rule a.e. Indeed, both factors are bounded there, and the increment of their product is bounded by a constant times the sum of their increments; this proves absolute

continuity from the definition. At the points where both derivatives exist, the ordinary product rule applies. Consequently

$$\int_a^b f'g = [fg]_a^b - \int_a^b fg'.$$

Likewise, the chain rule gives $(f^2)' = 2ff'$ a.e. For an infinite interval, first use these identities on $[0, T]$ and then justify the limit using integrability and the endpoint limits. This order of operations is used in the manuscript's quadratic completion of squares.

Absolute continuity cannot be replaced by mere continuity plus existence of a derivative a.e. The Cantor function is continuous, has derivative zero a.e., and has different endpoint values. Its negative therefore has derivative $0 \geq 0$ a.e. while its value drops from 0 to -1 on $[0, 1]$. The missing hypothesis is precisely the integral representation in Theorem 2.11.

2.6 Total variation and the cost of reversing direction

For a function on $[a, b]$, its total variation is

$$\mathrm{Var}_{[a,b]}(f) = \sup_{a=t_0<\cdots<t_n=b} \sum_{j=1}^{n} |f(t_j) - f(t_{j-1})|.$$

The supremum is over all finite partitions. A function is of bounded variation when this quantity is finite.

Proposition 2.13 (Variation of an absolutely continuous function)**.** *If f is absolutely continuous on $[a, b]$, then*

$$\mathrm{Var}_{[a,b]}(f) = \int_a^b |f'(t)|\,\mathrm{d}t.$$

Proof. For every partition, each increment is the integral of $g = f'$ over its subinterval. The triangle inequality gives the upper bound. For the converse, approximate g in $L^1(a, b)$ by an interval step function h with $\|g - h\|_1 < \varepsilon$. Such step functions are dense in L^1 on a bounded interval, a standard consequence of Lebesgue approximation [12]. Choose a partition on whose subintervals h is constant. Then

$$\sum_j \left| \int_{t_{j-1}}^{t_j} g \right| \geq \sum_j \left| \int_{t_{j-1}}^{t_j} h \right| - \varepsilon = \int_a^b |h| - \varepsilon \geq \int_a^b |g| - 2\varepsilon.$$

Take the supremum and let $\varepsilon \downarrow 0$. □

If y is locally absolutely continuous, $y' \in L^1$, $y(0) = 0$, and $y(\infty) = 1$, Proposition 2.2 applied to y' gives

$$\mathcal{V}_y := \int_0^\infty |y'| = 1 + 2B_y, \qquad B_y := \int_0^\infty (-y')_+. \tag{2.4}$$

The quantity B_y is the total distance travelled backwards. If $y(t) > 1$, the later motion must undo at least $y(t) - 1$, since

$$y(t) - 1 = -\int_t^\infty y'(s)\,\mathrm{d}s \leq \int_t^\infty (-y'(s))_+\,\mathrm{d}s \leq B_y.$$

Taking a supremum gives the peak-overshoot estimate $P_y \leq B_y$. These statements contain the real-analysis part of manuscript Theorem 3; the plant balance supplies the remaining bound on B_y.

Example 2.14 (An integrable signal that keeps returning to one)**.** Let $\varepsilon_n = 2^{-n-3}$ and

$$f(t) = \sum_{n=1}^{\infty} \max\left\{1 - \frac{|t-n|}{\varepsilon_n}, 0\right\}.$$

The triangular supports are disjoint, and only finitely many meet a bounded interval. Thus f is continuous and locally absolutely continuous. Its integral is $\sum_n \varepsilon_n = 1/8$, but $f(n) = 1$ and $f(n+1/2) = 0$ for every positive integer n. It has no limit at infinity. Indeed, even $\int e^{\eta t} f(t)\,\mathrm{d}t < \infty$ for $0 < \eta < \log 2$, because the contribution of the nth triangle is at most $e^{\eta(n+\varepsilon_n)}\varepsilon_n$. The total variation is infinite: each triangle contributes two. Thus the condition $f' \in L^1$ in Corollary 2.12 has real content. The example also explains why weighted integrability alone is insufficient for a pointwise exponential-decay claim.

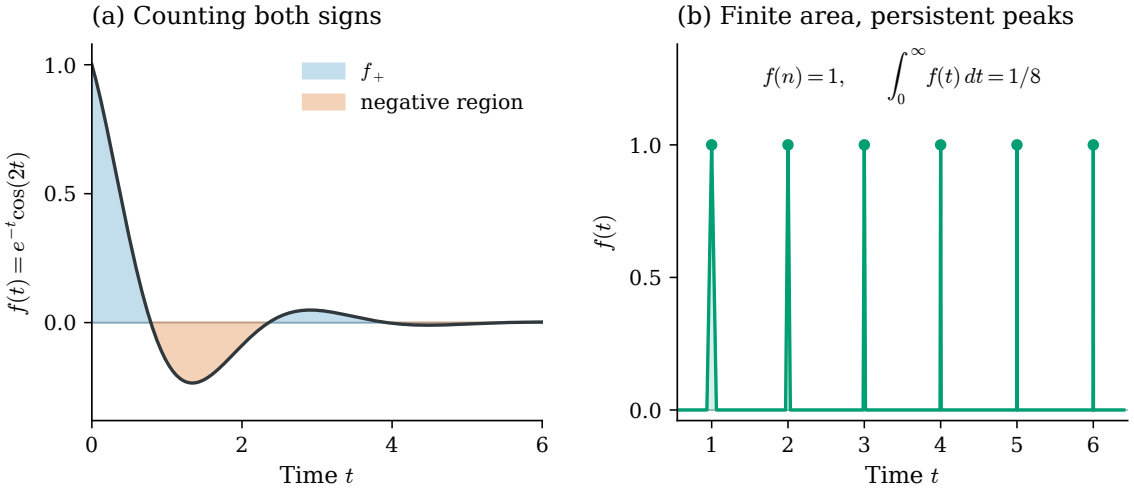


Figure 2.1: Two analytical examples. Left: positive and negative regions of $e^{-t}\cos(2t)$ contribute with the same sign to its absolute area. Right: the first six triangles in Example 2.14. The infinite family has finite area while its values at positive integers remain one. These signals illustrate integration facts; no plant model is assigned to them.

2.7 The strict inequality at the attainment boundary

Suppose $v \geq 0$, $0 < \int v < \infty$, and its exponential moment $J_v = \int_0^\infty e^t v(t)\,\mathrm{d}t$ is finite. Since $m(\{0\}) = 0$, a positive amount of ordinary function area must lie at times $t > 0$. On that set $e^t - 1 > 0$, so Lemma 2.1 gives

$$\int v < \int e^t v = J_v. \tag{2.5}$$

Manuscript Theorem 15 first uses Fatou's lemma to establish $J_v \leq 1$. To see the precise pattern, suppose $g_T \geq 0$, $\int g_T \leq 1$, and $g_T(r) \to e^r v(r)$ for each fixed r. Apply Fatou along any sequence $T_n \to \infty$ to get $J_v \leq 1$. Equation (2.5) then makes $\int v < 1$. This is the measure-theoretic reason that the weight condition in that theorem is strict. A mass concentrated at $t = 0$ would evade the argument; an ordinary integrable command cannot do so.

2.8 Two exercises with solutions

Exercise 2.15 (From area equality to a monotone output)**.** Let e be locally absolutely continuous, $e(0) = 1$, and $e, v \in L^1(\mathbb{R}_+)$ satisfy $e' = -e - v$ a.e. Suppose $\int |e| + \int |v| = 1$. Show that $e, v \geq 0$ a.e. and that $y = 1 - e$ is nondecreasing from zero to one.

Solution. Since $e' \in L^1$, Corollary 2.12 gives $e(\infty) = 0$. Integrating the equation gives $\int(e+v) = 1$. Subtract this from the assumed absolute-area equality and use (2.1): $2\int(e_- + v_-) = 0$. Both negative parts vanish a.e. Continuity gives $e \geq 0$ everywhere and the plant equation gives $y' = e + v \geq 0$ a.e. Absolute continuity makes y nondecreasing; its endpoints are 0 and 1. □

Exercise 2.16 (An ordinary command approaching a point mass)**.** For $0 < a < 1$, put $v_a(t) = (1-a)e^{-t/a}/a$. Compute its ordinary area, exponential moment, and initial value. Explain why no member has area one even though the areas approach one as $a \downarrow 0$.

Solution. Direct integration gives

$$\int_0^\infty v_a = 1 - a, \qquad \int_0^\infty e^t v_a(t)\,\mathrm{d}t = \frac{1-a}{a}\frac{1}{1/a - 1} = 1, \qquad v_a(0) = \frac{1-a}{a}.$$

The initial value diverges as $a \downarrow 0$. Each ordinary command has area strictly less than its exponential moment by (2.5). Thus the limiting area is approached through increasingly concentrated commands, while the endpoint is excluded from the ordinary-function class. This is the family used in the manuscript's complete-attainment construction. □

Chapter 3

Convolution, positivity, and stable inversion

A useful way to read the later proofs is to begin with a simple equation,

$$z = g + K * z.$$

The forcing g is known. The kernel K describes how an earlier value of z contributes to its present value. If this contribution is sufficiently small in an integral norm, repeated substitution produces a convergent series. If every term is nonnegative, the solution is nonnegative as well. These two observations account for much of manuscript Lemma 9 and Theorem 17. The same argument, with signed impulse measures, underlies the robustness estimate in manuscript Theorem 23.

Throughout this chapter, functions are defined on $[0, \infty)$ and are extended by zero to negative times when they enter a convolution. Equality between elements of L^p means equality almost everywhere. The normed-space background can be read in [11]; the integral inequalities use the Tonelli–Fubini facts in [12]. The convolution-resolvent viewpoint is developed systematically in [7].

3.1 Why completeness is needed

Definition 3.1 (Norm, completeness, and a bounded operator)**.** A norm on a real or complex vector space X is a function $x \mapsto \|x\|$ such that $\|x\| = 0$ exactly when $x = 0$, $\|ax\| = |a|\,\|x\|$, and $\|x + y\| \le \|x\| + \|y\|$. A sequence is Cauchy if its terms become arbitrarily close in this norm. A *Banach space* is a normed space in which every Cauchy sequence has a limit in the space. A linear map $T : X \to Y$ is bounded if

$$\|T\| := \sup_{\|x\| \le 1} \|Tx\| < \infty.$$

Then $\|Tx\| \le \|T\|\,\|x\|$, and T is continuous.

The word "bounded" here concerns the size of an operator relative to the input norm. A bounded operator can act on functions whose pointwise values are unbounded. Conversely, a bounded function need not have a finite L^1 norm on the infinite interval.

Theorem 3.2 (L^1 completeness through summable series)**.** *If $f_n \in L^1$ and $\sum_{n=1}^{\infty} \|f_n\|_1 < \infty$, then $\sum_{n=1}^{\infty} f_n$ converges almost everywhere and in L^1. In particular, $L^1(0, \infty)$ is a Banach space.*

Proof. Tonelli's theorem gives

$$\int_0^{\infty} \sum_{n=1}^{\infty} |f_n(t)|\,\mathrm{d}t = \sum_{n=1}^{\infty} \|f_n\|_1 < \infty.$$

The nonnegative sum is therefore finite almost everywhere. Define $f(t) = \sum_n f_n(t)$ on this full-measure set and put $f = 0$ on the exceptional set. The sum is measurable, and

$$\left\| f - \sum_{n=1}^{N} f_n \right\|_1 \le \sum_{n>N} \|f_n\|_1 \longrightarrow 0.$$

This also proves that $f \in L^1$.

Now let (u_n) be Cauchy in L^1. Choose a subsequence (u_{n_j}) with $\left\|u_{n_{j+1}} - u_{n_j}\right\|_1 \le 2^{-j}$. The series

$$u_{n_1} + \sum_{j=1}^{\infty} (u_{n_{j+1}} - u_{n_j})$$

converges in L^1 by the first part, so this subsequence has an L^1 limit u. The Cauchy property and the triangle inequality show that the whole sequence tends to u. □

The analogous completeness theorem holds for every L^p, with $1 \le p \le \infty$. The proofs in the manuscript mainly require the L^1 result just established. The general L^p theorem is standard background; it is not needed to justify an unmentioned change of norm.

Lemma 3.3 (The positive cone is closed)**.** *If $f_n \ge 0$ almost everywhere and $f_n \to f$ in L^1, then $f \ge 0$ almost everywhere.*

Proof. The negative part $f_- := \max(-f, 0)$ satisfies $f_- \le |f - f_n|$ almost everywhere. Consequently $\|f_-\|_1 \le \|f - f_n\|_1 \to 0$. A nonnegative function with zero integral vanishes almost everywhere. □

This small lemma explains why positivity survives a limit of partial sums. Norm convergence does not itself give pointwise convergence of the entire sequence, and that stronger assertion is unnecessary here.

3.2 Convolution as an operator

For causal functions define

$$(K * f)(t) = \int_0^t K(t-s) f(s)\, \mathrm{d}s. \tag{3.1}$$

The integral is defined for almost every t under the hypotheses below. Writing it over the whole real line after zero extension gives the same object. Associativity and commutativity follow from changes of variables and Fubini whenever the absolute integrals are finite.

Theorem 3.4 (The two convolution estimates used in the paper)**.** *For $K, f \in L^1$,*

$$\|K * f\|_1 \le \|K\|_1 \, \|f\|_1 .$$

For $K \in L^1$ and $f \in L^\infty$,

$$\|K * f\|_\infty \le \|K\|_1 \, \|f\|_\infty .$$

If both factors are nonnegative, their convolution is nonnegative, and

$$\int_0^\infty (K * f)(t)\, \mathrm{d}t = \left(\int_0^\infty K(t)\, \mathrm{d}t \right) \left(\int_0^\infty f(t)\, \mathrm{d}t \right). \tag{3.2}$$

Proof. Apply Tonelli to the absolute values and put $r = t - s$:

$$\begin{aligned} \int_0^\infty |(K * f)(t)|\, \mathrm{d}t &\le \int_0^\infty \int_0^t |K(t-s)|\, |f(s)|\, \mathrm{d}s\, \mathrm{d}t \\ &= \int_0^\infty |f(s)| \int_0^\infty |K(r)|\, \mathrm{d}r\, \mathrm{d}s. \end{aligned}$$

This proves the first estimate and almost-everywhere existence. For the second, bound $|f(s)|$ by its essential supremum inside the integral. Nonnegative factors give a nonnegative integrand. With those signs, the same change of variables proves the mass identity without the triangle inequality. More generally the signed mass identity holds for arbitrary L^1 factors, by Fubini. □

Thus $T_K f = K * f$ is a bounded operator on L^1 with $\|T_K\| \le \|K\|_1$. It is positive when $K \ge 0$: a nonnegative input produces a nonnegative output. A positive kernel of mass one is an L^1 contraction. This is the precise reason that adding the remaining normalized positive lags cannot enlarge the delay-error estimate in manuscript Theorem 23.

Example 3.5 (A cascade of two normalized lags)**.** Let $h_a(t) = ae^{-at}$ and $h_b(t) = be^{-bt}$, where $a, b > 0$. Each kernel is nonnegative and has mass one. Their convolution is

$$(h_a * h_b)(t) = \begin{cases} \dfrac{ab}{b-a}(e^{-at} - e^{-bt}), & a \ne b, \\ a^2 t e^{-at}, & a = b. \end{cases}$$

Both expressions are nonnegative and have mass one. Positivity and mass already followed from the convolution theorem; the formulas are useful for understanding repeated poles. In particular, the apparent singularity at $a = b$ belongs to the partial-fraction expression, not to the cascade response.

3.3 Finite measures include feedthrough and exact delay

There is no integrable function that acts as the identity for convolution. The identity is the Dirac measure δ_0, defined by $\delta_0(E) = 1$ if $0 \in E$ and 0 otherwise. More generally, $\int \phi(t)\, d\delta_T(t) = \phi(T)$ for a bounded Borel function ϕ. Hence

$$\delta_0 * f = f, \qquad (\delta_T * f)(t) = \mathbf{1}_{[T,\infty)}(t) f(t - T).$$

The first formula represents direct feedthrough; the second represents transport delay.

Definition 3.6 (Total variation of a finite signed measure)**.** For a finite signed Borel measure ν on $[0, \infty)$, its total variation on a Borel set E is

$$|\nu|(E) = \sup \left\{ \sum_{j=1}^{m} |\nu(E_j)| : E = \bigcup_{j=1}^{m} E_j, \quad E_j \text{ disjoint Borel sets} \right\}.$$

Its norm is $\|\nu\|_{\mathrm{TV}} = |\nu|([0, \infty))$. The variation $|\nu|$ is a positive measure, and signed integration satisfies

$$\left| \int \phi \, d\nu \right| \le \int |\phi| \, d|\nu|.$$

These are the standard total-variation facts for signed measures.

For the measures used in the paper, this abstract definition has a particularly simple form. If distinct times $T_1, \dots, T_m$ carry atoms and $k \in L^1$, then

$$d\nu = \sum_{j=1}^{m} a_j \, d\delta_{T_j} + k(t)\, \mathrm{d}t, \qquad \|\nu\|_{\mathrm{TV}} = \sum_{j=1}^{m} |a_j| + \|k\|_1 . \tag{3.3}$$

Atoms have zero Lebesgue measure, so their variation and that of the density are supported on disjoint sets. Coefficients of atoms at the same time must first be added.

For finite signed measures ν, μ, define convolution by

$$(\nu * \mu)(E) = \int\int \mathbf{1}_E(s + t)\, d\nu(s)\, d\mu(t).$$

For a function, choose a Borel measurable representative and set $(\nu * f)(t) = \int_{[0,t]} f(t - s)\, d\nu(s)$ wherever the integral exists. The estimates below define an L^1 class independently of that choice: Tonelli's theorem shows that changing f on a Lebesgue null set changes the convolution only on a Lebesgue null set. Pointwise bounds refer to the chosen representative. These definitions agree with ordinary convolution for measures with integrable densities.

Theorem 3.7 (The measure convolution algebra)**.** *Finite signed Borel measures, with convolution and the total-variation norm, form a commutative Banach algebra with identity δ_0. Moreover,*

$$\|\nu * \mu\|_{\mathrm{TV}} \le \|\nu\|_{\mathrm{TV}} \, \|\mu\|_{\mathrm{TV}}, \qquad \|\nu * f\|_1 \le \|\nu\|_{\mathrm{TV}} \, \|f\|_1 .$$

The corresponding L^∞ estimate also holds.

Proof. For each finite Borel partition (E_j) of $[0,\infty)$,

$$\begin{aligned}\sum_j |(\nu * \mu)(E_j)| &\le \int\int \sum_j \mathbf{1}_{E_j}(s+t)\, d|\nu|(s)\, d|\mu|(t) \\ &= \|\nu\|_{\mathrm{TV}} \, \|\mu\|_{\mathrm{TV}} .\end{aligned}$$

Taking the supremum proves the first estimate. Tonelli applied to $|f(t-s)|\, d|\nu|(s)\, \mathrm{d}t$ proves the second exactly as in Theorem 3.4; the supremum estimate is obtained in the same way. Fubini proves associativity and commutativity, and the defining property of δ_0 gives the identity.

For completeness, consider an absolutely summable series of measures, $\sum_n \|\nu_n\|_{\mathrm{TV}} < \infty$. Define $\nu(E) = \sum_n \nu_n(E)$. If (E_j) is a disjoint sequence, then

$$\sum_n \sum_j |\nu_n(E_j)| \le \sum_n \|\nu_n\|_{\mathrm{TV}} < \infty.$$

Interchanging the two sums proves countable additivity of ν. The variation norm of each series tail is bounded by the sum of the tail norms. Thus the series converges in total variation. The same Cauchy-subsequence argument used in Theorem 3.2 proves completeness. □

Example 3.8 (The two filters in the robustness theorem)**.** For $0 < a < 1$,

$$F_a(p) = \frac{1+p}{1+ap} = \frac{1}{a} + \frac{1-1/a}{1+ap}.$$

Its impulse measure is

$$d\nu_a = \frac{1}{a}\, d\delta_0 - \frac{1-a}{a^2} e^{-t/a}\, \mathrm{d}t, \qquad \|\nu_a\|_{\mathrm{TV}} = \frac{2}{a} - 1.$$

The finite-delay filter $F_{c,T}(p) = 1 + c - ce^{-Tp}$ has impulse measure $(1+c)\delta_0 - c\delta_T$. For $c \ge 0$ and $T > 0$, its norm is $1 + 2c$. At $T = 0$ the atoms combine and the norm is 1.

Although these impulse responses have atoms, their responses to a step are ordinary functions with finite jumps. The exclusion of an impulsive physical command in the manuscript therefore does not exclude these filters. The signal and the impulse response of the operator acting on that signal are different mathematical objects.

3.4 The geometric series becomes an inverse

Theorem 3.9 (Neumann inversion)**.** *Let $\mathcal{A}$ be a Banach algebra with identity I of norm one, and let $K \in \mathcal{A}$ satisfy $q = \|K\| < 1$. Then*

$$(I-K)^{-1} = \sum_{n=0}^{\infty} K^n, \qquad \left\|(I-K)^{-1}\right\| \le \frac{1}{1-q}. \tag{3.4}$$

The partial sum through $n = N$ has error at most $q^{N+1}/(1-q)$. Replacing K by $-K$ gives the inverse of $I + K$.

Proof. Submultiplicativity gives $\|K^n\| \le q^n$. Completeness therefore gives a limit $R = \sum_{n\ge 0} K^n$. The finite identity

$$(I-K)\sum_{n=0}^{N} K^n = \left(\sum_{n=0}^{N} K^n\right)(I-K) = I - K^{N+1}$$

passes to the limit because multiplication is continuous and $\left\|K^{N+1}\right\| \to 0$. This proves both inverse identities. Summing the norm bounds gives the two estimates. □

The algebra may consist of bounded operators on a Banach space, or of finite measures under convolution. In the latter case $I = \delta_0$ and K^n means the n-fold convolution K^{*n}.

Corollary 3.10 (A positive resolvent)**.** *Let $K \ge 0$ belong to L^1, with $q = \|K\|_1 < 1$. For every $g \in L^1$, the equation $z = g + K * z$ has the unique L^1 solution*

$$z = g + K * g + K^{*2} * g + \cdots, \qquad \|z\|_1 \le \frac{\|g\|_1}{1-q}.$$

If $g \ge 0$, then $z \ge 0$ almost everywhere. In this case its mass is exactly $\int z = (\int g)/(1-q)$.

Proof. The series converges by $\|K^{*n} * g\|_1 \le q^n \|g\|_1$. Theorem 3.9 proves the equation. If z_1, z_2 were two solutions, Young's inequality would give $\|z_1 - z_2\|_1 \le q\,\|z_1 - z_2\|_1$, forcing equality of the solutions. Positivity follows from Lemma 3.3. For nonnegative g, use the convolution mass identity and sum the scalar geometric series. □

This is manuscript Lemma 9 with its steps made explicit. Positivity is an additional conclusion of the nonnegative series. A Neumann inverse with alternating signed terms still has a norm bound, but that bound alone does not imply a nonnegative response.

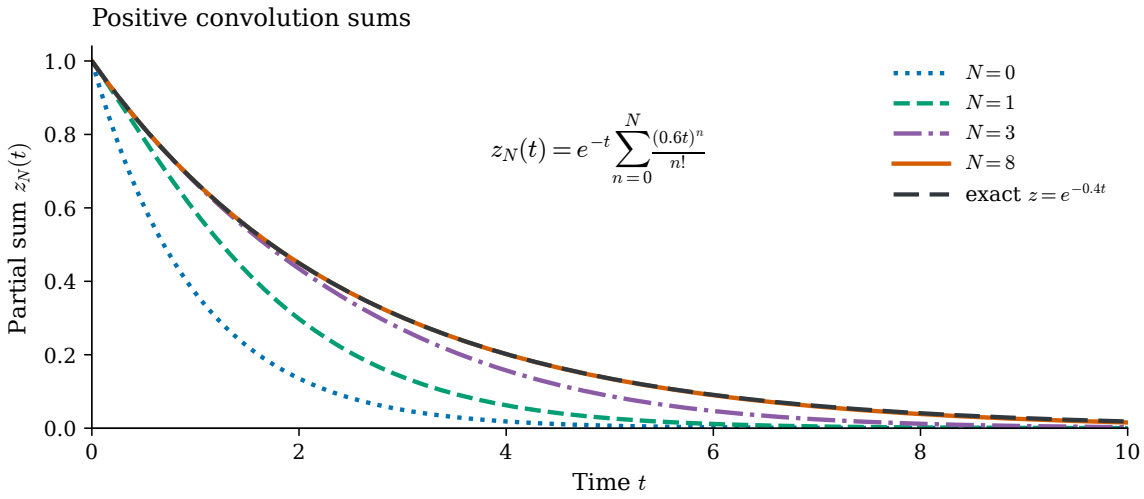


Figure 3.1: Repeated convolution builds the positive resolvent. For $K(t) = 0.6e^{-t}$ and $g(t) = e^{-t}$, the partial sums increase towards $z(t) = e^{-0.4t}$. The uniform geometric bound on the sum of the L^1 norms justifies the limiting operation.

3.5 Exponential weights and actual decay

An ordinary integral norm controls total area. To retain information about decay, define, for $\eta \ge 0$,

$$\|f\|_{1,\eta} = \int_0^\infty e^{\eta t}|f(t)|\,\mathrm{d}t, \qquad \|\nu\|_{\mathrm{TV},\eta} = \int_{[0,\infty)} e^{\eta t}\,d|\nu|(t).$$

Multiplication by $e^{\eta t}$ identifies each weighted space isometrically with its unweighted version. For example,

$$e^{\eta t}(K * f)(t) = \big[(e^{\eta \cdot} K) * (e^{\eta \cdot} f)\big](t).$$

The same factorization holds for measures, because $e^{\eta(s+t)} = e^{\eta s} e^{\eta t}$. Consequently the weighted convolution inequalities, completeness, and Neumann theorem all follow from their unweighted versions.

Lemma 3.11 (A strict norm bound persists under a small weight)**.** *Suppose* $\|\nu\|_{\mathrm{TV}} < 1$ *and* $\|\nu\|_{\mathrm{TV},\eta_0} < \infty$ *for some* $\eta_0 > 0$*. Then* $\|\nu\|_{\mathrm{TV},\eta} < 1$ *for every sufficiently small* $\eta > 0$*.*

Proof. As $\eta \downarrow 0$, the integrand $e^{\eta t}$ tends to one and is bounded by $e^{\eta_0 t}$ for $0 \le \eta \le \eta_0$. Dominated convergence with respect to $|\nu|$ gives $\|\nu\|_{\mathrm{TV},\eta} \to \|\nu\|_{\mathrm{TV}} < 1$. □

The strict inequality matters. An unweighted norm equal to one leaves no margin for this argument. A finite positive exponential moment is also a genuine hypothesis; ordinary integrability does not supply it.

Proposition 3.12 (A weighted supremum estimate)**.** *Suppose* $|g(t)| \le Ce^{-\eta t}$ *for every* $t \ge 0$ *and* $\|\nu\|_{\mathrm{TV},\eta} < \infty$*. The convolution defined by these representatives satisfies*

$$|(\nu * g)(t)| \le C \|\nu\|_{\mathrm{TV},\eta} e^{-\eta t} \qquad (t \ge 0).$$

If $\nu = \delta_0 + \sum_{n\ge 1} K^{*n}$ *and* $q_\eta = \|K\|_{1,\eta} < 1$*, the factor on the right is at most* $C/(1 - q_\eta)$*.*

Proof. Use the pointwise bound at $t - s$:

$$e^{\eta t}|(\nu * g)(t)| \le C \int_{[0,t]} e^{\eta s}\, d|\nu|(s) \le C \|\nu\|_{\mathrm{TV},\eta}.$$

The final assertion is the weighted Neumann estimate. □

An essential-supremum formulation gives the same bound almost everywhere for L^∞ classes. A continuous response then satisfies the bound everywhere. Manuscript Theorem 17 uses an explicit exponentially bounded base kernel and its convolution representative, so the estimate gives actual pointwise decay. Finite weighted L^1 norm alone would not justify that conclusion: continuous spikes can become narrower while retaining large peak values.

For internal stability, these estimates must apply to the internal dynamics of the stated realization. The manuscript checks that connection through its characteristic factorization or stable-block realization. A bound for one input-output response alone does not complete that separate step.

3.6 Why a small delay becomes small after a lag

Example 3.13 (Two different distances between delays)**.** For $d > 0$, the atoms at 0 and d are disjoint, and hence

$$\|\delta_d - \delta_0\|_{\mathrm{TV}} = 2.$$

This distance does not tend to zero with d. Now take the ordinary lag kernel $g(t) = e^{-t}\mathbf{1}_{[0,\infty)}(t)$ and its delayed version $g_d(t) = \mathbf{1}_{[d,\infty)}(t)e^{-(t-d)}$. Before d, only g is present; after d, $g_d \ge g$. Therefore

$$\begin{aligned}\|g - g_d\|_1 &= \int_0^d e^{-t}\,\mathrm{d}t + \int_d^\infty (e^{-(t-d)} - e^{-t})\,\mathrm{d}t \\ &= 2(1 - e^{-d}) \longrightarrow 0.\end{aligned}$$

Convolution with any additional nonnegative unit-mass kernel cannot increase this distance. This is the smoothing step in the delay-error bound of manuscript Theorem 23.

There is a related area identity for step responses. Let $h \geq 0$ have mass one and put $S(t) = \int_0^t h(s)\,\mathrm{d}s$, with $S = 0$ at negative times. Since S is nondecreasing, $S(t) - S(t-d) \geq 0$. Tonelli gives

$$\int_0^\infty |S(t) - S(t-d)|\,\mathrm{d}t = \int_0^\infty h(s) \int_s^{s+d} \mathrm{d}t\,\mathrm{d}s = d. \tag{3.5}$$

Notice that S itself is not in L^1: it tends to one. Its difference from the delayed step response is in L^1, with the exact area above.

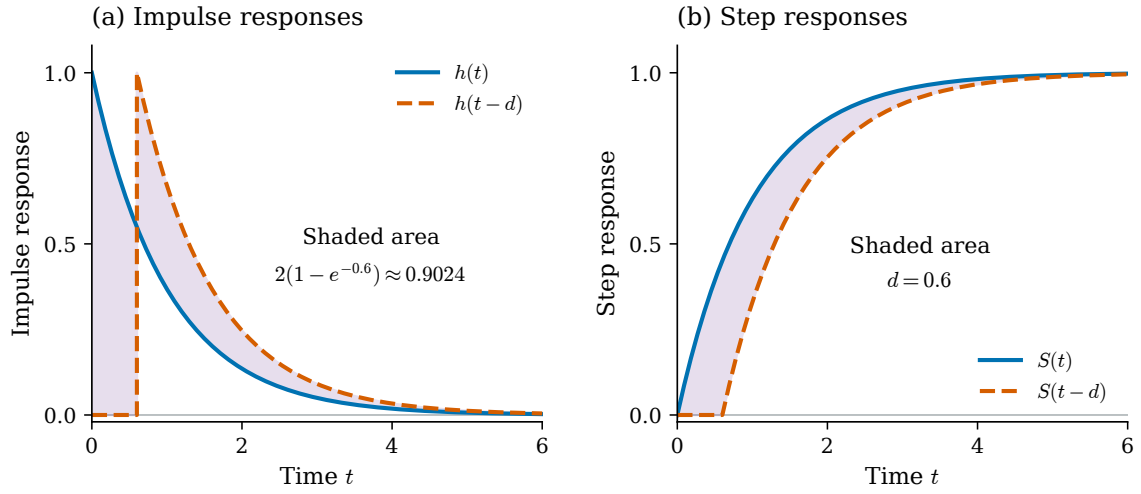


Figure 3.2: An ordinary lag smooths a delay perturbation. At $d = 0.6$, the absolute difference of the two impulse responses has integral $2(1 - e^{-d})$, while the difference of the two step responses has integral d. These are the two estimates used in the delay-robustness argument.

3.7 From signal perturbations to MIACE perturbations

Proposition 3.14 (A product norm and its continuity estimate)**.** *For $\alpha > 0$, define on $L^1 \times L^1$*

$$\|(e, v)\|_\alpha = \sqrt{\|e\|_1^2 + \alpha\,\|v\|_1^2}.$$

This is a norm and makes the product a Banach space. For two signal pairs,

$$|\|(e, v)\|_\alpha - \|(\widetilde{e}, \widetilde{v})\|_\alpha| \leq \sqrt{\|e - \widetilde{e}\|_1^2 + \alpha\,\|v - \widetilde{v}\|_1^2}. \tag{3.6}$$

Proof. Definiteness and homogeneity follow from those of L^1. For the triangle inequality, first use the L^1 triangle inequality separately in the two components, then the Euclidean triangle inequality in $\mathbb{R}^2$. Completeness follows componentwise, or from

$$\|(e, v)\|_\alpha \leq \|e\|_1 + \sqrt{\alpha}\,\|v\|_1 \leq \sqrt{2}\,\|(e, v)\|_\alpha\,.$$

The reverse triangle inequality for this product norm proves (3.6). □

This proof uses the Euclidean norm only after taking the two signal L^1 norms. It does not turn L^1 into a Hilbert space. The estimate is exactly suited to the last step of manuscript Theorem 23. If its previous calculations give

$$\|e_\varepsilon - e_0\|_1 \leq \frac{2fd}{1 - q_d}, \qquad \|v_\varepsilon - v_0\|_1 \leq \frac{f^2 d}{1 - q_d},$$

then substitution gives

$$|\mathcal{M}_{\alpha,\varepsilon} - \mathcal{M}_{\alpha,0}| \le \frac{fd\sqrt{4+\alpha f^2}}{1-q_d}.$$

The centered commands must use the same equilibrium for this comparison. In the manuscript's delay perturbation, the DC gain is unchanged, so $v_\varepsilon - v_0 = w_\varepsilon - w_0$.

3.8 Two exercises with solutions

Exercise 3.15 (Summing a resolvent explicitly)**.** Let $K(t) = be^{-at}$, where $0 \le b < a$, and let $g(t) = e^{-at}$. Find the solution of $z = g + K * z$ directly from the convolution series. Find its L^1 norm and the exponential weights for which the series converges in weighted L^1.

Solution. Induction by direct integration gives $K^{*n} * g = b^n t^n e^{-at}/n!$ for $n \ge 0$, where the $n = 0$ term is g. Thus

$$z(t) = e^{-at} \sum_{n=0}^{\infty} \frac{(bt)^n}{n!} = e^{-(a-b)t}, \qquad \|z\|_1 = \frac{1}{a-b}.$$

For $0 \le \eta < a$, the weighted kernel norm is $b/(a-\eta)$. If $b > 0$, it is below one exactly when $\eta < a - b$, which is also the exact range of weighted integrability of z. If $b = 0$, the equation is $z = g$ and the same range $\eta < a$ follows from the forcing alone. □

Exercise 3.16 (Checking the small-gain margin)**.** Use the filter F_a with $a = 1/2$ in manuscript Theorem 23. Determine the delay-error range supplied by the norm estimate and evaluate the MIACE deviation bound for $\alpha = 1$ and $d = 0.02$.

Solution. Example 3.8 gives $f = 2/a - 1 = 3$. Hence $q_d = 6(1 - e^{-d}) < 1$ exactly when

$$d < \log(6/5) \simeq 0.18232.$$

At $d = 0.02$, $q_d \simeq 0.118808$, and the bound is

$$\frac{0.06\sqrt{13}}{1 - 6(1 - e^{-0.02})} \simeq 0.2455.$$

This is an absolute bound on the change of normalized MIACE supplied by the estimate. It is not an assertion that the actual change equals this value, or that the perturbed response remains monotone. The calculation illustrates how positivity, contraction, and norm continuity produce a quantitative sufficient result. □

Chapter 4

Laplace transforms, analytic continuation, and exact delay remainders

A transform identity initially obtained for positive real frequencies can contain information at negative frequencies. The passage is justified by decay of the signal and analyticity of its transform. This is how the paper proves that a controller attaining the area bound must cancel a particular plant pole. Later, interpolation at the plant poles produces an exact decomposition of the delay. The remainder remains in the calculation.

This chapter supplies the ingredients for manuscript Theorem 7, Proposition 8, Lemma 16, and the constructions that follow them. Time is normalized and denoted by t; the complex transform variable is p.

4.1 Existence and the elementary transform rules

Definition 4.1 (One-sided Laplace transform)**.** For a locally integrable function $f : [0, \infty) \to \mathbb{C}$, define

$$\widehat{f}(p) = \int_0^\infty e^{-pt} f(t)\, \mathrm{d}t$$

whenever the integral is absolutely convergent. In transform manipulations a causal function is extended by zero to negative times, unless a different physical prehistory is explicitly specified.

If $f \in L^1(0, \infty)$, the integral exists for $\operatorname{Re} p \geq 0$, and $|\widehat{f}(p)| \leq \|f\|_1$. A stronger weighted estimate will be used below. For a real a with $e^{-at} f(t) \in L^1$, it gives

$$|\widehat{f}(p)| \leq \int_0^\infty e^{-at} |f(t)|\, \mathrm{d}t, \qquad \operatorname{Re} p \geq a.$$

Proposition 4.2 (Value at zero)**.** *If $f \in L^1(0, \infty)$, then*

$$\lim_{p \downarrow 0} \widehat{f}(p) = \widehat{f}(0) = \int_0^\infty f(t)\, \mathrm{d}t,$$

where p approaches zero through positive real values.

Proof. The factors e^{-pt} tend to one and satisfy $|e^{-pt} f(t)| \leq |f(t)|$. Dominated convergence applies. The statement is about a signed integral when f is real. It identifies an absolute area only after nonnegativity has been established. □

Proposition 4.3 (Derivative, causal shift, and convolution)**.** *The following identities hold under the indicated assumptions.*

1. *Suppose f is locally absolutely continuous, $f(0)$ is its right-hand value, $e^{-pt} f$ and $e^{-pt} f'$ are integrable, and $e^{-pT} f(T) \to 0$. Then*

$$\widehat{f'}(p) = p\widehat{f}(p) - f(0).$$

2. *For $d \geq 0$, set $f_d(t) = \mathbf{1}_{[d,\infty)}(t) f(t - d)$. Whenever $\widehat{f}(p)$ exists,*

$$\widehat{f_d}(p) = e^{-dp} \widehat{f}(p).$$

3. If $e^{-at}f, e^{-at}g \in L^1(0,\infty)$ for some real a, then for $\operatorname{Re} p \geq a$,

$$\widehat{f * g}(p) = \widehat{f}(p)\widehat{g}(p), \qquad (f * g)(t) = \int_0^t f(t-s)g(s)\,\mathrm{d}s.$$

Proof. For the first identity, integrate by parts on $[0,T]$ and use the stated boundary limit. The second follows by setting $s = t - d$. For the third, the double integral of the absolute value is bounded by

$$\int_0^\infty \int_0^t e^{-at}|f(t-s)||g(s)|\,\mathrm{d}s\,\mathrm{d}t = \left\|e^{-a\cdot}f\right\|_1 \left\|e^{-a\cdot}g\right\|_1 < \infty.$$

Fubini's theorem therefore permits the substitution $r = t - s$ and the separation into the two transform integrals. □

The derivative formula requires its boundary and integrability assumptions. In the paper they follow from the stable differential equations or from explicit exponential estimates. Iterating it gives, for example, $\widehat{f''} = p^2\widehat{f} - pf(0) - f'(0)$ when the corresponding hypotheses hold.

Example 4.4 (Centering changes the prehistory)**.** Suppose the physical scaled command w is zero before the step, and put $v = w - 1$. Its physical prehistory is $v = -1$. Write v_c for the causal extension of its values at nonnegative times. Then, for $t \geq 0$,

$$v(t-d) = v_\mathrm{c}(t-d) - \mathbf{1}_{[0,d)}(t),$$

up to irrelevant endpoint values. Hence, for $\operatorname{Re} p > 0$,

$$\mathcal{L}\{v(\cdot - d)\}(p) = e^{-dp}\widehat{v}(p) - \frac{1 - e^{-dp}}{p}.$$

Applying the zero-prehistory shift rule directly to the physical centered command would omit the last term. In the delayed first-order plant, setting $f(t) = e(t+d)$ instead gives $f' = -f - v$ and $f(0) = 1$. The waiting interval then contributes exactly d to the absolute error area.

4.2 Exponential moments and holomorphic transforms

Theorem 4.5 (Analyticity from an exponential moment)**.** *Suppose $\int_0^\infty e^{\eta t}|f(t)|\,\mathrm{d}t < \infty$ for a real η. Then $\widehat{f}$ is holomorphic on $\operatorname{Re} p > -\eta$, and*

$$\widehat{f}^{(n)}(p) = \int_0^\infty (-t)^n e^{-pt} f(t)\,\mathrm{d}t, \qquad n = 0, 1, 2, \ldots.$$

If f has compact support, its transform is entire.

Proof. On a compact subset of the stated half-plane choose $\varepsilon > 0$ so that $\operatorname{Re} p + \eta \geq 2\varepsilon$. For each fixed n,

$$t^n|e^{-pt}f(t)| \leq (t^n e^{-2\varepsilon t})e^{\eta t}|f(t)| \leq C_{n,\varepsilon}e^{\eta t}|f(t)|.$$

The right side is integrable. For a sufficiently small complex increment h, the difference quotient of e^{-pt} is bounded by $te^{-(\operatorname{Re} p - |h|)t}$; the same estimate with a smaller ε dominates it. Dominated convergence proves complex differentiability under the integral. Repetition proves the formula for every n. A compactly supported integrable function has all finite exponential moments, so its transform is analytic on every such half-plane. □

There is a useful boundary distinction. A bound $|f(t)| \leq Ce^{-bt}$ implies analyticity for $\operatorname{Re} p > -b$, because every weight $\eta < b$ is integrable. It need not imply integrability at weight b. For example, the transform of e^{-bt} is $(p+b)^{-1}$ and has a pole at the boundary $p = -b$.

Theorem 4.6 (Uniqueness of the Laplace transform: standard background)**.** *Let f, g be locally integrable and let $e^{-at}f, e^{-at}g \in L^1$ for some real a. If $\widehat{f}(p) = \widehat{g}(p)$ for every real $p > a$, then $f = g$ almost everywhere.*

This standard uniqueness theorem can be obtained from uniqueness of the Fourier transform on L^1: analyticity first extends the equality to the half-plane; on a line $p = b + \mathrm{i}\omega$, $b > a$, the transforms are the Fourier transforms of the zero-extended functions $e^{-bt}f(t)$ and $e^{-bt}g(t)$. Fourier uniqueness is the background input in this route. The paper uses the resulting uniqueness principle, rather than a general inverse-transform contour theorem; see [7].

Proposition 4.7 (Positivity on the real transform axis)**.** *Suppose $f \geq 0$ almost everywhere, f is not zero almost everywhere, and $\widehat{f}(a)$ is finite for a real a. Then $\widehat{f}(a) > 0$. If $a \leq 0$ and $f \in L^1$, then also*

$$\widehat{f}(a) \geq \widehat{f}(0) > 0.$$

Proof. The weight e^{-at} is strictly positive. Thus its product with f is positive on a set of positive measure. The second assertion follows from $e^{-at} \geq 1$ for $a \leq 0$. □

4.3 The small part of complex analysis that is needed

A holomorphic function is complex differentiable in an open set. Its local Taylor expansion is the relevant fact here. A function is meromorphic when its singularities in that set are poles. If F has a pole of order q at z, its Laurent expansion has the form

$$F(p) = \sum_{j=1}^{q} \frac{c_j}{(p-z)^j} + H(p),$$

where H is holomorphic near z. The finite sum is its *principal part*; the coefficient c_1 is its residue. A removable singularity is filled by a holomorphic value. These concepts and the following theorem are standard elementary complex analysis [10].

Theorem 4.8 (Identity theorem)**.** *Two holomorphic functions on a connected open set that agree on a subset with an accumulation point inside the set agree everywhere in the set.*

Proof from local Taylor expansions. Subtract the functions. If the resulting function is not identically zero near a zero z, take the first nonzero Taylor coefficient there. Locally the function is $(p-z)^q H(p)$ with $H(z) \neq 0$, so z is isolated among its zeros. An interior accumulation of zeros therefore forces vanishing on a neighborhood. The set of points having a neighborhood of vanishing is open. At a limit point inside the domain the same Taylor argument shows that it is also closed. Connectedness makes it the whole domain. □

To apply this theorem to a quotient, first multiply by the denominator. The resulting product identity is analytic even where the displayed denominator might vanish. Its zeros can then be examined without assuming in advance that the quotient is defined there.

Example 4.9 (Why equality forces the PI cancellation)**.** For positive PI parameters x, r, the manuscript obtains

$$\widehat{E}(p) = \frac{r(p+1)}{D(p)}, \qquad D(p) = rp(p+1) + x(rp+1)e^{-dp}.$$

Suppose equality in the MIACE area bound has already given $e, v \geq 0$. For $f(t) = e(t+d)$ the plant equation gives $f' \leq -f$, $f(0) = 1$, so $0 \leq f(t) \leq e^{-t}$. The error equals one on the waiting interval and has this exponential bound afterwards. Therefore $\widehat{E}$ is analytic on $\operatorname{Re} p > -1$.

The identity $D(p)\widehat{E}(p) = r(p+1)$ originally holds in a right half-plane. The identity theorem extends it to $\operatorname{Re} p > -1$. At $p = -a$, $0 < a < 1$, the right side is nonzero, so $D(-a) \neq 0$. If $r \neq 1$, then as $a \uparrow 1$,

$$D(-a) \longrightarrow x(1-r)e^{d} \neq 0, \qquad r(1-a) \longrightarrow 0.$$

The quotient would give $\widehat{E}(-a) \to 0$. Positivity instead gives

$$\widehat{E}(-a) = \int_0^\infty e^{at} e(t)\,\mathrm{d}t \geq \widehat{E}(0) = r/x > 0.$$

This contradiction proves $r = 1$. The cancellation follows from attainment of the lower bound. It is not an assumption inserted into the minimization. The same reasoning, applied to a different numerator and denominator, underlies manuscript Proposition 8 for filtered PID.

4.3.1 Repeated poles and exponential polynomials

For $\operatorname{Re} p > \operatorname{Re} z$ and an integer $j \geq 1$,

$$\mathcal{L}\left\{ \frac{t^{j-1}}{(j-1)!} e^{zt} \right\}(p) = \frac{1}{(p-z)^j}. \tag{4.1}$$

For $j = 1$ this is the integral of an exponential. Integration by parts proves the general formula inductively. Thus the inverse transform of a strictly proper rational function is a sum of terms $t^k e^{zt}$, with the exponents and polynomial degrees determined by its poles and their multiplicities. A repeated pole requires these polynomial factors; treating it as a collection of distinct simple poles can lead to singular-looking coefficients although the signal itself has a regular limit.

The exponential polynomial has an analytic continuation to negative t. That continuation is a mathematical function used in the next construction. The actual causal impulse response still vanishes at negative time.

4.4 Hermite interpolation and the simplex formula

Proposition 4.10 (Confluent Hermite interpolation)**.** *Let $z_1, \ldots, z_s$ be distinct nodes with positive multiplicities $q_1, \ldots, q_s$, and let $m = \sum_j q_j$. Prescribe the values of a polynomial and its first $q_j - 1$ derivatives at each z_j. There is a unique polynomial I of degree less than m satisfying these m conditions.*

Proof. The map taking a polynomial of degree less than m to the prescribed derivative values is a linear map between two m-dimensional spaces. A polynomial in its kernel is divisible by $\prod_j (p - z_j)^{q_j}$, of degree m, so it must be zero. Injectivity in equal finite dimensions gives bijectivity. This proves existence as well as uniqueness. □

Repeated nodes therefore mean matching derivatives. For the delay function, the conditions are $I^{(\ell)}(z) = (-d)^\ell e^{-dz}$. They ensure that $e^{-dp} - I(p)$ has a zero of the required multiplicity at each plant pole. Equivalently, $e^{-dp}/B(p)$ and $I(p)/B(p)$ have the same principal parts at the roots of B. Divided differences and this confluent construction are discussed in [3].

Proposition 4.11 (Simplex representation for a cascade kernel)**.** *Let $m \geq 1$, $r_j > 0$, and*

$$B(p) = \prod_{j=1}^{m} (1 + p/r_j).$$

The causal impulse response h_B of $1/B$ satisfies, for $t > 0$,

$$h_B(t) = \left(\prod_{j=1}^{m} r_j \right) t^{m-1} \int_{\Delta_{m-1}} \exp\left(-t \sum_{j=1}^{m} r_j u_j \right) \mathrm{d}u. \tag{4.2}$$

Here $u_j \geq 0$, $\sum_j u_j = 1$, *and* $\mathrm{d}u$ *means ordinary coordinate measure in* $u_1, \ldots, u_{m-1}$, *with* $u_m = 1 - \sum_{j<m} u_j$. *Its total mass is* $1/(m-1)!$. *For* $m = 1$, *the integral means evaluation at* $u_1 = 1$.

Proof. Each factor has impulse response $r_j e^{-r_j t}$ on $t \geq 0$. Successive convolution writes their product transform as an integral over $s_j \geq 0$, $\sum_j s_j = t$. In the independent coordinates $s_1, \ldots, s_{m-1}$, substitute $s_j = t u_j$. The Jacobian is t^{m-1}, giving (4.2). No separation of the rates was used, so equal rates are included. □

The right side extends to an entire function of t: the simplex is compact and the integrand is an exponential with bounded rate. It agrees with the exponential-polynomial continuation of h_B for $t > 0$, hence everywhere by the identity theorem. At a negative real time the integral is strictly positive; only t^{m-1} changes sign. Therefore

$$\operatorname{sign} h_B(t) = (-1)^{m-1}, \qquad t < 0.$$

This short argument supplies the sign needed in the all-order construction, including repeated lags. It is the relevant instance of the Hermite–Genocchi integral representation [3].

4.5 An exact compactly supported remainder

Theorem 4.12 (Delay decomposition used in manuscript Lemma 16)**.** *Let* B *be as above, let* $d \geq 0$, *and let* I_d *be the degree-less-than-m Hermite interpolant of* e^{-dp} *at the roots of* B, *with their multiplicities. Define*

$$J(t) = (-1)^{m-1} h_B(t-d) \quad (0 \leq t \leq d), \qquad J(t) = 0 \quad (t > d).$$

The values at the two endpoints do not affect its transform. Then $J \geq 0$ *almost everywhere, and for every complex* p,

$$e^{-dp} = I_d(p) + (-1)^m B(p) \widehat{J}(p). \tag{4.3}$$

Proof. The sign follows from (4.2). The rational function I_d/B is strictly proper. Its principal parts are those of e^{-dp}/B at every root z of B. To identify its inverse transform, multiply the principal part at z by e^{pt} and take the coefficient of $(p-z)^{-1}$. Formula (4.1) shows that summing these coefficients over the poles gives the inverse of I_d/B. The same coefficients can be computed from $e^{p(t-d)}/B(p)$, and hence give the exponential-polynomial function $h_B(t-d)$.

The causal inverse of e^{-dp}/B equals zero for $t < d$ and equals $h_B(t-d)$ for $t > d$. Subtracting the inverse of I_d/B, the difference is $-h_B(t-d) = (-1)^m J(t)$ on $0 < t < d$ and zero afterwards. Taking Laplace transforms in a right half-plane and multiplying by B proves (4.3) there. Every term in that identity is entire: J has compact support, so $\widehat{J}$ is entire by Theorem 4.5. The identity theorem extends the equality to all p, including the roots of B. □

The polynomial and the remainder together equal the delay exactly. Dropping $B\widehat{J}$ would create an approximation; the manuscript keeps this term and uses its nonnegative kernel to prove stability and positivity.

Example 4.13 (One lag: the sign of the remainder)**.** Take $m = 1$, $B(p) = 1 + p$. Then $h_B(t) = e^{-t}$, $I_d(p) = e^d$, and $J(t) = e^{d-t}$ on $[0, d]$. Direct integration gives

$$\widehat{J}(p) = e^d \frac{1 - e^{-(p+1)d}}{p+1}, \qquad e^{-dp} = e^d - (1+p)\widehat{J}(p).$$

The quotient for $\widehat{J}$ is interpreted by continuity at $p = -1$, where its value is de^d. The apparent pole is removable.

Example 4.14 (Two equal lags: a repeated interpolation node)**.** Take $m = 2$ and $B(p) = (1+p)^2$. Then $h_B(t) = te^{-t}$ and

$$I_d(p) = e^d\big[1 - d(p+1)\big], \qquad J(t) = (d-t)e^{d-t} \quad (0 \le t \le d).$$

Indeed, $I_d(-1) = e^d$ and $I_d'(-1) = -de^d$, the two Hermite conditions. With $q = p+1$, integration yields

$$\widehat{J}(p) = e^d \frac{dq - 1 + e^{-dq}}{q^2}, \qquad I_d(p) + (1+p)^2\widehat{J}(p) = e^{-dp}.$$

The numerator in the quotient is $d^2q^2/2 + O(q^3)$, so $\widehat{J}(-1) = e^d d^2/2$. The positive sign of the remainder here and the negative sign in the preceding example are exactly the factor $(-1)^m$.

4.5.1 Ordinary moments and exponential moments

For a nonnegative integrable J supported in $[0, d]$, set

$$j_0 = \int_0^d J(t)\,\mathrm{d}t, \qquad j_1 = \int_0^d tJ(t)\,\mathrm{d}t, \qquad \Phi(\lambda) = \int_0^d e^{\lambda t}J(t)\,\mathrm{d}t.$$

Then $j_0 = \widehat{J}(0)$, $j_1 = -\widehat{J}'(0)$, and $\Phi(\lambda) = \widehat{J}(-\lambda)$. The first two are ordinary moments; the third is an exponential moment. They answer different questions in the stability estimates.

Define $K(t) = \int_t^d J(r)\,\mathrm{d}r$ for $0 \le t \le d$ and set it to zero afterwards. Tonelli gives

$$\int_0^\infty K(t)\,\mathrm{d}t = \int_0^d \int_0^r J(r)\,\mathrm{d}t\,\mathrm{d}r = j_1.$$

Fubini, applicable on this compact integration region, gives for $p \neq 0$

$$\widehat{K}(p) = \int_0^d J(r) \int_0^r e^{-pt}\,\mathrm{d}t\,\mathrm{d}r = \frac{j_0 - \widehat{J}(p)}{p}.$$

At $p = 0$, the quotient has the removable value j_1. These identities explain the tail kernel and the quantities appearing in manuscript Theorem 17.

4.6 Exercises with solutions

Exercise 4.15 (A transform with a repeated pole)**.** Let $f(t) = te^{-2t}$ for $t \ge 0$. Find its Laplace transform, its absolute area, and its first ordinary moment. Determine the open half-plane of analyticity and decide whether the exponential moment at weight 2 is finite.

Solution. Formula (4.1) gives $\widehat{f}(p) = (p+2)^{-2}$ for $\operatorname{Re} p > -2$. Since $f \ge 0$, its absolute area is $\widehat{f}(0) = 1/4$. Differentiating gives $-\widehat{f}'(0) = 2/2^3 = 1/4$ for the first moment. At weight 2, the integral becomes $\int_0^\infty t\,\mathrm{d}t = \infty$. The open analyticity half-plane therefore does not include its boundary. □

Exercise 4.16 (The remainder for a lag of arbitrary length)**.** For $B(p) = 1 + \sigma p$, $\sigma > 0$, find I_d and J in (4.3). Verify the identity directly and compute j_0 and j_1.

Solution. Here $h_B(t) = \sigma^{-1}e^{-t/\sigma}$ and the only interpolation node is $-1/\sigma$. Thus

$$I_d = e^{d/\sigma}, \qquad J(t) = \sigma^{-1}e^{(d-t)/\sigma}\mathbf{1}_{[0,d]}(t).$$

Direct integration yields

$$\widehat{J}(p) = \frac{e^{d/\sigma}\big(1 - e^{-(p+1/\sigma)d}\big)}{1 + \sigma p}.$$

Consequently $I_d - B\widehat{J} = e^{-dp}$, with the quotient extended at its removable singularity. The two moments are

$$j_0 = e^{d/\sigma} - 1, \qquad j_1 = \sigma(e^{d/\sigma} - 1) - d.$$

For $d = 0$ both vanish. For positive d, nonnegativity is also immediate from their defining integrals. □

Chapter 5

Differential equations, delay, and quadratic energy

An integral inequality becomes a statement about a controlled system only after the differential equation has been used. The same two absolute areas can belong to different time histories. The plant balance determines which histories are possible, and stability determines whether their infinite tails can be integrated. This chapter collects the dynamical facts needed for those steps in the manuscript.

The finite-dimensional ODE and energy arguments below are proved directly. Three delay-equation results are identified as classical inputs: the characteristic-root criterion for a retarded system, the scalar stability threshold, and the scalar oscillation criterion. Their hypotheses are stated so that the reader can check exactly where they apply.

5.1 Linear ODEs with ordinary inputs

An input with a finite jump is still an ordinary locally integrable function. Its integral has no jump. This distinction explains why the ideal PI command can jump at the reference step while the first-order plant output remains continuous.

Proposition 5.1 (Variation of constants)**.** *Let $A \in \mathbb{R}^{n\times n}$, let $b \in L^1_{\mathrm{loc}}([0,\infty);\mathbb{R}^n)$, and let $x_0 \in \mathbb{R}^n$. The unique locally absolutely continuous solution of $x' = Ax + b$ almost everywhere, with $x(0) = x_0$, is*

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-s)}b(s)\,\mathrm{d}s.$$

Proof. The absolutely convergent series $e^{At} = \sum_{j=0}^{\infty} A^j t^j/j!$ can be differentiated on bounded time intervals and satisfies $(e^{At})' = Ae^{At}$. Multiplication of the equation by e^{-At} gives $(e^{-At}x(t))' = e^{-At}b(t)$ almost everywhere. The fundamental theorem for absolutely continuous functions then gives the formula. Conversely, that formula is locally absolutely continuous and satisfies the equation. The difference of two solutions satisfies the same calculation with zero input and zero initial value, so it vanishes. □

For a first-order lag, $\sigma x' + x = u$ with $\sigma > 0$, the formula is

$$x(t) = e^{-t/\sigma}x(0) + \frac{1}{\sigma}\int_0^t e^{-(t-s)/\sigma}u(s)\,\mathrm{d}s. \tag{5.1}$$

Its kernel is nonnegative and has integral one. Thus a nonnegative initial state and a nonnegative input give a nonnegative state. If $x(0) = 0$ and $0 \le u \le 1$, then $0 \le x(t) \le 1 - e^{-t/\sigma} \le 1$.

Proposition 5.2 (A lag preserves a nondecreasing input)**.** *Suppose u is nonnegative and nondecreasing on $[0,\infty)$ and the lag starts from zero. Then $0 \le x(t) \le u(t)$ at every continuity point of u, and x is nondecreasing. If $u(t) \to 1$, then $x(t) \to 1$.*

Proof. For $s \le t$, $u(s) \le u(t)$. Inserting this bound into (5.1) gives $x(t) \le u(t)(1 - e^{-t/\sigma}) \le u(t)$. Therefore $x' = (u - x)/\sigma \ge 0$ almost everywhere, and absolute continuity gives monotonicity. To obtain the limit, fix $\epsilon > 0$ and take T so that $1 - \epsilon \le u(t) \le 1 + \epsilon$ for $t \ge T$. Variation of constants from T bounds $x(t)$ between two solutions with constant inputs $1 - \epsilon$ and $1 + \epsilon$; their limits are those constants. Let ϵ tend to zero. □

Applying this argument successively proves the corresponding assertion for a cascade of stable real lags. A final transport delay preserves monotonicity and simply adds an initial interval on which the output is zero. This is the elementary positivity mechanism behind the explicit attaining output transfers in the manuscript.

5.2 Why the equality signs imply monotonicity in a cascade

The command itself need not be nondecreasing. At MIACE equality the relevant signs are $w \geq 1$ and $y \leq 1$. A different argument is therefore needed to handle the entire attaining command class.

Lemma 5.3 (One-way crossing of the target)**.** *Let $x_j(0) = 0$ and*

$$\sigma_j x_j' + x_j = x_{j-1}, \qquad j = 1, \dots, N, \qquad x_0 = w \geq 1 \quad \text{a.e.},$$

where every $\sigma_j > 0$ and w is locally integrable. Each state is nondecreasing until it first reaches one. After reaching one, it cannot return below one.

Proof. For the first state, $x_1' = (w - x_1)/\sigma_1 \geq 0$ while $x_1 \leq 1$. If $x_1(T) = 1$, variation of constants gives, for $t \geq T$,

$$x_1(t) = e^{-(t-T)/\sigma_1} + \frac{1}{\sigma_1} \int_T^t e^{-(t-s)/\sigma_1} w(s)\, \mathrm{d}s \geq 1.$$

Suppose the result holds for x_{j-1}. Before that state reaches one, it is nonnegative and nondecreasing, so the lag formula gives $x_j(t) \leq x_{j-1}(t)$ and hence $x_j' \geq 0$. If x_{j-1} reaches one, it subsequently stays at least one. On that later interval, $x_j' \geq 0$ whenever $x_j \leq 1$, and the same variation-of-constants calculation prevents a downward crossing after x_j reaches one. Induction proves the assertion. □

If the last state also satisfies $x_N \leq 1$ at every time, it is nondecreasing throughout its motion. Thus the two equality signs in the MIACE bound supply exactly the assumptions needed for the manuscript's Theorem 14. The claim uses the cascade structure: an arbitrary stable transfer function need not have this crossing property.

5.3 Exponential stability and the role of the realization

Definition 5.4. A real or complex square matrix is *Hurwitz* if every eigenvalue has strictly negative real part. The homogeneous ODE $x' = Ax$ is exponentially stable if there are constants $M \geq 1$ and $\gamma > 0$ such that $\left\| e^{At} \right\| \leq M e^{-\gamma t}$ for $t \geq 0$.

Proposition 5.5 (Hurwitz matrices and decay)**.** *A matrix is Hurwitz if and only if its homogeneous ODE is exponentially stable.*

Proof. Over $\mathbb{C}$, write $A = SJS^{-1}$ in Jordan form. On a Jordan block $J_0 = \lambda I + N$, with $N^m = 0$,

$$e^{J_0 t} = e^{\lambda t} \left(I + tN + \cdots + \frac{t^{m-1} N^{m-1}}{(m-1)!} \right).$$

If all real parts are negative, choose $\gamma > 0$ smaller than their absolute distances from the imaginary axis. Each polynomial factor is bounded after multiplication by the remaining decaying exponential. Multiplication by S and S^{-1} gives the required estimate. Conversely, an eigenvector with eigenvalue λ evolves as $e^{\lambda t}$ times that vector. An exponential decay estimate forces $\operatorname{Re} \lambda < 0$. □

Repeated poles therefore produce factors such as $te^{-\lambda t}$, without destroying exponential decay when $\lambda > 0$. This observation justifies many integrability assumptions in the delay-free PI calculations.

A rational transfer function is *proper* if its numerator degree does not exceed its denominator degree, and *strictly proper* if the inequality is strict. In a state-space realization,

$$x' = Ax + Bu, \qquad y = Cx + Du, \qquad H(p) = C(pI - A)^{-1}B + D.$$

The constant D is direct feedthrough. Its impulse response is a Dirac mass, whereas $Ce^{At}B$ is an ordinary function for $t \geq 0$.

Example 5.6 (A stable transfer can hide an unstable state)**.** Take $A = \operatorname{diag}(-1, 1)$, $B = (1, 0)^{\mathsf{T}}$, and $C = (1, 0)$. The zero-state transfer is $1/(p+1)$, yet the unobserved second state equals $e^t x_2(0)$. The input-output map is stable and the specified realization is not internally stable. This is why a cancelled factor must be checked in the retained state equations.

For the manuscript's cancellation family, a cancelled plant pole at -1 is stable. A complete proof must still establish stability of all remaining modes. The PID and arbitrary-order constructions do this by factoring the full characteristic determinant of a physical realization, and then using a positive convolution resolvent for the remaining factor.

5.4 Retarded delay equations and the method of steps

Consider the constant-coefficient system

$$X'(t) = A_0 X(t) + \sum_{j=1}^{m} A_j X(t - \tau_j), \qquad \tau_j > 0, \tag{5.2}$$

with a continuous initial history on $[-\tau_{max}, 0]$. It is called *retarded* because delayed states appear but delayed derivatives do not. On the first time interval, every delayed term is known from the history; an ODE determines the new state. Repeating this procedure gives the method of steps and a unique continuous solution, which is locally absolutely continuous for $t \geq 0$.

Substitution of $X(t) = e^{pt}v$ gives the characteristic matrix

$$D(p) = pI - A_0 - \sum_{j=1}^{m} A_j e^{-p\tau_j}.$$

Its determinant is generally not a polynomial and can have infinitely many zeros.

Theorem 5.7 (Classical characteristic-root criterion)**.** *For the finite-dimensional retarded system* (5.2)*, exponential stability on the continuous-history space is equivalent to*

$$s_* := \sup\{\operatorname{Re} p : \det D(p) = 0\} < 0.$$

Here exponential stability means that constants $M, \gamma > 0$ bound the solution history at time t by $Me^{-\gamma t}$ times the norm of its initial history.

This is a classical input from retarded functional differential equations; its general proof is not reproduced here. The resolvent or solution semigroup theory that proves it is developed in [8]. The assumptions exclude neutral equations with delayed derivatives, for which the same argument cannot simply be reused.

In the manuscript the application is concrete. A characteristic factor has the form

$$D_0(p)[1 - \widehat{K}(p)],$$

where the zeros of D_0 lie strictly to the left and $\int_0^\infty e^{\eta t}|K(t)|\,\mathrm{d}t < 1$ for some $\eta > 0$. For $\operatorname{Re} p \geq -\eta$,

$$|\widehat{K}(p)| \leq \int_0^\infty e^{\eta t}|K(t)|\,\mathrm{d}t < 1.$$

Thus the second factor cannot vanish in this strip. Once all internal factors have been accounted for, Theorem 5.7 turns this explicit estimate into internal exponential stability.

5.4.1 The scalar fundamental solution

For $x > 0$ and $\delta > 0$, let

$$g'(t) = -xg(t-\delta), \qquad g(0) = 1, \qquad g(t) = 0 \quad (t < 0). \tag{5.3}$$

The jump at zero is the convention for a fundamental solution. The equation is interpreted almost everywhere for $t > 0$. It gives $g = 1$ on $[0, \delta]$, followed by $g(t) = 1 - x(t-\delta)$ on $[\delta, 2\delta]$. Continuing yields the locally finite formula

$$g(t) = 1 + \sum_{j=1}^{\lfloor t/\delta \rfloor} \frac{(-x)^j (t - j\delta)^j}{j!}, \qquad t \geq 0. \tag{5.4}$$

At a step boundary the new summand vanishes, so the formula is continuous. Termwise differentiation away from boundaries verifies the equation, and uniqueness by the method of steps proves the formula everywhere. For $\delta = 0$, the solution is $g(t) = e^{-xt}$ instead.

Although the fundamental history has a jump at zero, the segment $g = 1$ on $[0, \delta]$ is continuous. Restarting the equation at time δ brings this solution within the continuous-history setting of Theorem 5.7. In the PI cancellation problem, the step error is precisely this g.

Theorem 5.8 (Classical scalar stability threshold)**.** *The scalar retarded equation $z'(t) = -xz(t-\delta)$, with $x > 0$, is exponentially stable exactly when $x\delta < \pi/2$.*

The full root classification is the classical scalar result associated with [9]. The boundary can be understood directly. Put $a = x\delta$ and $z = \delta p$; the characteristic equation is $z + ae^{-z} = 0$. If $z = u + iv$ with $u \geq 0$, then

$$u = -ae^{-u}\cos v, \qquad v = ae^{-u}\sin v.$$

When $a < \pi/2$, $|v| \leq a < \pi/2$, so the first equation gives $u < 0$, a contradiction. Roots in any fixed right half-strip are bounded, since $|z| = ae^{-u}$, which also excludes an accumulation toward the imaginary axis without an imaginary root. This proves the stable side. An imaginary root first occurs at $z = i\pi/2$, $a = \pi/2$. Implicit differentiation gives

$$\frac{\mathrm{d}z}{\mathrm{d}a} = \frac{z}{a(1+z)}, \qquad \operatorname{Re}\frac{\mathrm{d}z}{\mathrm{d}a} = \frac{v^2}{a(1+v^2)} > 0 \quad (z = iv).$$

Thus this pair crosses into the right half-plane. The classical root-count argument completes the global converse. At the boundary, the imaginary roots exclude exponential decay.

5.4.2 Positivity has a stricter threshold

Proposition 5.9 (A direct positivity proof)**.** *If $x\delta \leq 1/e$, the fundamental solution in (5.3) is positive at every finite nonnegative time and is nonincreasing. It tends to zero exponentially.*

Proof. The function $\lambda e^{-\lambda\delta}$ has maximum $1/(e\delta)$. Hence there is $\lambda > 0$ satisfying $\lambda = xe^{\lambda\delta}$. Set $h(t) = e^{\lambda t}g(t)$, with $h(t) = 0$ for $t < 0$. The equation becomes

$$h' = \lambda[h(t) - h(t-\delta)].$$

Integrating the derivative of the moving-window expression gives

$$h(t) = 1 + \lambda \int_{\max(0,t-\delta)}^{t} h(s)\,\mathrm{d}s. \tag{5.5}$$

If h had a first zero, the right side would be at least one at that time, a contradiction. Thus $g > 0$. Equation (5.3) then gives $g' \leq 0$. Since $1/e < \pi/2$, exponential decay follows from Theorem 5.8. □

For the converse, the classical constant-coefficient oscillation result says that, if $x\delta > 1/e$, every real nontrivial solution of the scalar equation is oscillatory in the eventual-sign sense: it is neither eventually strictly positive nor eventually strictly negative. The continuous-delay statement is recalled in the introduction, equations (17)–(20), of [2]; the positivity framework is also discussed in [4]. This is a classical input, separate from the direct sufficiency proof above. A globally nonnegative fundamental solution would be nonincreasing; a first zero would have a strictly negative derivative because its delayed value is still positive. It would therefore stay strictly positive at every finite time. Applied after the initial waiting interval, the oscillation criterion rules this out above $1/e$.

The two thresholds therefore answer different questions. Between $1/e$ and $\pi/2$, the transient decays but eventually crosses the target. On the cancellation family the centered command satisfies $v(t) = xg(t) - g(t+\delta)$. If $x \geq 1$ and g is nonnegative and nonincreasing, then $v(t) \geq (x-1)g(t) \geq 0$. These are the sign conditions used in the manuscript's Lemma 6 and Theorem 7.

Example 5.10 (A delay small enough for stability can still permit overshoot)**.** For $x = 2$ and $\delta = 0.2$, the product is 0.4. It lies between $1/e$ and $\pi/2$. The fundamental error therefore decays exponentially but cannot stay nonnegative. For $x = 2$ and $\delta = 0.1$, the product is $0.2 < 1/e$, and the output $1 - g$ is monotone. This comparison uses the exact delay equation, with no polynomial approximation of the delay.

5.5 Stable internal-model realizations

Let P be the manuscript's finite cascade of positive first-order lags with a finite causal transport delay. Let F be a proper filter realized by Hurwitz finite-dimensional states and finite causal delay blocks. These assumptions include the filters F_a and $F_{c,T}$ used in the manuscript. Retain those stable realizations, with bounded initial delay histories. Consider

$$\widehat{y} = Pw, \qquad w = F(r - y + \widehat{y}).$$

The variable $\widehat{y}$ is the output of an actual retained plant model. With matching parameters and zero initial conditions, $y = Pw$, so $w = Fr$ and $y = PFr$. The equivalent error-feedback controller is $C = F/(1 - PF)$. This is the standard internal-model construction [6].

For internal stability, retain initial conditions. The difference between the plant and model states evolves autonomously because both are driven by the same command. Stable lag states decay exponentially; a difference in a finite transport-line history leaves after a finite time. The stable filter is therefore driven by the reference plus a decaying difference. Its homogeneous response, the resulting command difference, and all plant and model states decay exponentially. A convolution of two exponentially decaying functions can contain a factor t, but it still decays exponentially at any slightly smaller rate.

This argument explains why the nominal simplification $Y/R = PF$ is consistent with internal stability in the specified realization. It also shows what must be reconsidered when the plant and model delays differ: the model-state difference is then coupled back into the command. The small-gain estimate of manuscript Theorem 23 controls that coupling.

5.6 Quadratic energy as an inner-product calculation

For real functions $f, g \in L^2(0,\infty)$, the bound $2|fg| \leq f^2 + g^2$ first gives $fg \in L^1$. Define $\langle f, g\rangle = \int_0^\infty fg$. The Cauchy–Schwarz inequality states

$$|\langle f, g\rangle| \leq \|f\|_2 \, \|g\|_2 .$$

To see it, expand the nonnegative quadratic $\|f - tg\|_2^2$ in the real parameter t and minimize it. If $g \neq 0$, the minimum occurs at $t = \langle f, g\rangle / \|g\|_2^2$; its nonnegativity gives the inequality. The case $g = 0$ is immediate. In particular the product of two L^2 functions is integrable.

Proposition 5.11 (The scalar quadratic lower bound)**.** *Suppose e is locally absolutely continuous, $e' = -e - v$ almost everywhere, $e(0) = 1$, $e(t) \to 0$, and $e, v \in L^2(0,\infty)$. For $q > 0$, put $s = \sqrt{1+q}$ and $J_q = q\,\|e\|_2^2 + \|v\|_2^2$. Then*

$$J_q - (s-1) = \int_0^\infty [v - (s-1)e]^2 \,\mathrm{d}t \geq 0. \tag{5.6}$$

Equality holds exactly when $e(t) = e^{-st}$ and $v(t) = (s-1)e^{-st}$, with equality of commands interpreted almost everywhere.

Proof. The differential equation gives $e' \in L^2$, so Cauchy–Schwarz makes ee' and ev integrable. On each finite interval the chain rule for absolutely continuous functions gives $\int_0^T ee' = (e(T)^2 - 1)/2$. Letting $T \to \infty$ and using the plant equation yields

$$\int_0^\infty ev \,\mathrm{d}t = \frac{1}{2} - \|e\|_2^2 .$$

Expand the right side of (5.6). The coefficient of $\|e\|_2^2$ becomes $2(s-1) + (s-1)^2 = s^2 - 1 = q$, and the constant is $-(s-1)$. Equality forces $v = (s-1)e$ almost everywhere. Therefore $e' = -se$, and the initial value determines the displayed exponential. Conversely, those signals satisfy every assumption and attain equality. □

The number $P = s - 1$ solves the scalar algebraic Riccati equation $P^2 + 2P - q = 0$. This is the elementary completion-of-squares form of the scalar LQR argument [1]. The manuscript then identifies the PI parameters producing these signals. The infinite-horizon endpoint assumption is used explicitly; finite energy alone is not a license to substitute a pointwise limit for an arbitrary signal.

5.7 Gramians and the second-order energy formula

Proposition 5.12 (The controllability Gramian)**.** *Let A be a real Hurwitz matrix and B a real matrix of compatible size. Then*

$$P = \int_0^\infty e^{At} B B^{\mathsf{T}} e^{A^{\mathsf{T}} t} \,\mathrm{d}t$$

exists entrywise, is positive semidefinite, and is the unique solution of

$$AP + PA^{\mathsf{T}} + BB^{\mathsf{T}} = 0.$$

For the scalar strictly proper transfer $H(p) = C(pI - A)^{-1}B$,

$$\int_0^\infty |Ce^{At}B|^2 \,\mathrm{d}t = CPC^{\mathsf{T}}.$$

Proof. The exponential estimate in Proposition 5.5 makes every entry absolutely integrable. For any vector z, $z^{\mathsf{T}} P z = \int |B^{\mathsf{T}} e^{A^{\mathsf{T}} t} z|^2 \,\mathrm{d}t \geq 0$. Differentiate the integrand and integrate from zero to infinity. Its limit at infinity is zero and its value at zero is BB^{T}, which gives the Lyapunov equation. If D is the difference of two solutions, then $AD + DA^{\mathsf{T}} = 0$, so $e^{At} D e^{A^{\mathsf{T}} t}$ is constant. Its limit is zero; hence $D = 0$. Finally, multiplication by C and C^{T} inside the integral gives the energy formula. □

Example 5.13 (A formula valid also at repeated poles)**.** For $a, b, c > 0$ and real d, e_0, take

$$H(p) = \frac{dp + e_0}{ap^2 + bp + c}, \quad A = \begin{bmatrix} 0 & 1 \\ -c/a & -b/a \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/a \end{bmatrix}, \quad C = [e_0 \;\; d].$$

The matrix is Hurwitz because the two roots have negative real parts: for real roots their sum is $-b/a < 0$ and product $c/a > 0$, while a complex conjugate pair has real part $-b/(2a) < 0$. Write $P_{12} = P_{21}$. The Lyapunov equation gives

$$2P_{12} = 0, \qquad P_{22} - \frac{c}{a}P_{11} - \frac{b}{a}P_{12} = 0, \qquad -\frac{2c}{a}P_{12} - \frac{2b}{a}P_{22} + \frac{1}{a^2} = 0.$$

Thus $P_{11} = 1/(2bc)$, $P_{12} = 0$, and $P_{22} = 1/(2ab)$. Consequently

$$\|H\|_2^2 = CPC^{\mathsf{T}} = \frac{ae_0^2 + cd^2}{2abc}. \tag{5.7}$$

Here $\|H\|_2^2$ denotes the squared L^2 norm of the impulse response. Substitution of the error and command numerators gives the manuscript's Proposition 25. No separation into distinct poles was used, so the formula remains valid at a repeated pole.

5.8 A delayed energy calculation through autocorrelation

Let e be the fundamental solution (5.3) in the stable range $0 < x\delta < \pi/2$. Define

$$R(s) = \int_0^\infty e(t)e(t+s)\,\mathrm{d}t, \qquad s \geq 0, \qquad \mathcal{E}_\delta = R(0).$$

Both e and its almost-everywhere derivative have an exponential bound after a finite initial interval. For s in a bounded interval, the products needed below have a common integrable bound. Difference quotients can therefore be passed under the integral by dominated convergence; local absolute continuity handles the finitely many initial breakpoints. For $0 < s < \delta$, zero prehistory gives

$$R'(s) = -x\int_{\delta-s}^\infty e(t)e(t+s-\delta)\,\mathrm{d}t = -xR(\delta - s).$$

It follows that $R''(s) = -x^2R(s)$. The chain rule and decay give

$$R'(0^+) = \int_0^\infty ee'\,\mathrm{d}t = -\frac{1}{2}, \qquad R(\delta) = \frac{1}{2x}.$$

Solving the ordinary equation for R on $[0,\delta]$ yields

$$R(s) = \mathcal{E}_\delta\cos(xs) - \frac{\sin(xs)}{2x}, \qquad \mathcal{E}_\delta = \frac{1+\sin(x\delta)}{2x\cos(x\delta)}.$$

The denominator is positive in the stipulated stable interval.

For the centered PI command $v(t) = xe(t) - e(t+\delta)$, square the identity and integrate. Since $e = 1$ on the waiting interval, $\int_0^\infty e(t+\delta)^2\,\mathrm{d}t = \mathcal{E}_\delta - \delta$. The cross term is $R(\delta) = 1/(2x)$, so

$$\mathcal{U}_\delta = (x^2+1)\mathcal{E}_\delta - \delta - 1. \tag{5.8}$$

The limiting formulas at $\delta = 0$ follow continuously. This is the calculation used in manuscript Proposition 27: exponential stability supplies the domination, the delay equation supplies the correlation identity, and an ordinary second-order ODE finishes the integration.

5.9 Exercises with solutions

Exercise 5.14 (A bounded jump and a continuous plant state)**.** Let $\sigma x' + x = w$, $x(0) = 0$, and $w(t) = 1 + c\mathbf{1}_{[0,T)}(t)$, where $c, T, \sigma > 0$. Find $x(t)$ before and after T. Determine exactly when $x(t) \leq 1$ for all time. Express the condition as an exponential-moment inequality.

Solution. Before T, $x(t) = (1+c)(1-e^{-t/\sigma})$. Afterwards,

$$x(t) = 1 + [x(T) - 1]e^{-(t-T)/\sigma}.$$

The two expressions agree at T, even though w jumps there. Before T the output increases; after T it approaches one from the side determined by $x(T) - 1$. Therefore nonovershoot is equivalent to $(1+c)(1-e^{-T/\sigma}) \le 1$, or $c(e^{T/\sigma} - 1) \le 1$. The latter is exactly

$$\frac{1}{\sigma}\int_0^\infty e^{t/\sigma}[w(t) - 1]\,\mathrm{d}t \le 1.$$

At equality the output reaches one at T and remains there. □

Exercise 5.15 (Energy without partial fractions)**.** Find the impulse-response energy of $H(p) = 1/(p+1)^2$ in two ways: by direct integration and by the Gramian formula. Then apply Proposition 5.11 at $q = 3$ and compute both optimal signal energies.

Solution. The impulse response is te^{-t}, so two integrations by parts give $\int_0^\infty t^2 e^{-2t}\,\mathrm{d}t = 1/4$. In (5.7), use $a = c = 1$, $b = 2$, $d = 0$, $e_0 = 1$; the result is again $1/4$. For $q = 3$, $s = 2$ and the optimal signals are $e = v = e^{-2t}$. Their energies are both $1/4$, so $J_3 = 3/4 + 1/4 = 1 = s - 1$. The vanishing squared residual in (5.6) verifies the equality case directly. □

Chapter 6

From analytical tools to optimal controllers

The preceding chapters explain the integrals, transforms, and solutions that occur in the manuscript. Several short arguments connect those objects to an optimization theorem. They determine the equality cases, separate a lower bound from an attainable optimum, and show why a small cost change need not preserve a sign condition.

6.1 A balance law and a weighted distance

Suppose e is locally absolutely continuous, $e(0) = 1$, $e(t) \to 0$, and $e, v \in L^1(0,\infty)$ satisfy $e' = -e - v$ almost everywhere. Integrating on $[0,T]$ and then passing to the limit gives

$$\int_0^\infty e + \int_0^\infty v = 1. \tag{6.1}$$

The limit of e is a stated hypothesis here; it does not follow from $e \in L^1$ alone. For the stable PI responses it follows from the differential equation.

There is a useful version of the same calculation for a general plant. Let its normalized transfer function satisfy

$$P(p) = e^{-\delta p}H(p), \qquad H(0) = 1, \qquad H(p) = 1 - \mu p + o(p) \quad (p \downarrow 0).$$

With zero physical prehistory, write the command transform as $W(p) = 1/p + V(p)$, where $V = \widehat{v}$. Since $e = 1 - y$,

$$E(p) = \frac{1 - P(p)}{p} - P(p)V(p).$$

If $e, v \in L^1$, dominated convergence gives $E(p) \to \int e$ and $V(p) \to \int v$. Consequently

$$\int_0^\infty e + \int_0^\infty v = S, \qquad S = \mu + \delta. \tag{6.2}$$

For a cascade $H(p) = \prod_{j=1}^N (1 + \sigma_j p)^{-1}$, $\mu = \sum_j \sigma_j$. Thus the signed-area balance knows the sum of the lag time constants and the transport delay without requiring a partial-fraction expansion. This is the starting point of manuscript Theorem 12.

Theorem 6.1 (An exact decomposition of the weighted area distance)**.** *Let e, v be real integrable signals satisfying (6.2), with $S > 0$. Put*

$$A_e = \|e\|_1, \quad A_u = \|v\|_1, \quad N = \int e_- + \int v_-, \quad m_\alpha^2 = \frac{\alpha}{1+\alpha}.$$

For every $\alpha > 0$,

$$\mathcal{M}_\alpha^2 - S^2 m_\alpha^2 = 4m_\alpha^2 N(S+N) + \frac{(A_e - \alpha A_u)^2}{1+\alpha}. \tag{6.3}$$

Equality $\mathcal{M}_\alpha = S m_\alpha$ holds precisely when $e, v \geq 0$ almost everywhere and

$$A_e = \frac{\alpha S}{1+\alpha}, \qquad A_u = \frac{S}{1+\alpha}.$$

Proof. The positive-part identity gives $A_e + A_u = S + 2N$. Direct expansion gives

$$(1+\alpha)(A_e^2 + \alpha A_u^2) = \alpha(A_e + A_u)^2 + (A_e - \alpha A_u)^2.$$

Substitute $S + 2N$ and subtract αS^2. Both terms in (6.3) are nonnegative. Their simultaneous vanishing means $N = 0$ and $A_e = \alpha A_u$. The zero-integral lemma for nonnegative functions converts $N = 0$ into the two almost-everywhere sign conditions, and solving the two linear area equations gives the displayed values. □

This is the precise connection between the complex-plane motivation and the proof. The point $A_e + i\sqrt{\alpha}A_u$ has modulus $\mathcal{M}_\alpha$. Orthogonal coordinates in the real plane split its squared distance into a component normal to the line $A_e + A_u = S$ and a component along it. The map from signals to that point first uses two L^1 norms; it is not a time integral of a pointwise Euclidean norm.

In the first-order case, equality also gives $y' = e + v \geq 0$ almost everywhere. Absolute continuity then gives monotonicity at every time. For a general stable plant, the signed-area identity alone says nothing about this derivative. The positive-cascade argument in the manuscript supplies the additional dynamical step.

Example 6.2 (The optimum within the cancellation family)**.** For the delay-free first-order plant and $x \geq 1$, take

$$e(t) = e^{-xt}, \qquad v(t) = (x-1)e^{-xt}.$$

Then $A_e = 1/x$ and $A_u = 1 - 1/x$. The ratio $A_e = \alpha A_u$ selects $x = 1 + 1/\alpha$. At $\alpha = 1$, both areas are $1/2$, while the error settling below a tolerance $0 < \varepsilon < 1$ takes time $\log(1/\varepsilon)/2$. This time formula follows from the particular exponential trajectory; the two area values alone do not determine a settling time.

6.2 Efficiency, attainability, and uniqueness

A point (a, b) in a feasible objective set is *Pareto efficient* if no other feasible point has both coordinates no larger and at least one smaller. A minimizer of $a^2 + \alpha b^2$ over nonnegative objectives is efficient when $\alpha > 0$, since decreasing either coordinate strictly decreases the objective. This implication does not say that every efficient point in an arbitrary objective set can be obtained by varying a weight.

Three separate steps occur in an exact tuning proof:

1. Derive a lower bound for every admissible response.
2. Construct an admissible, internally stable controller attaining it.
3. Analyze the equality cases within the declared controller class.

The third step can yield a unique controller even though multiple signals outside that class have the same optimal areas. A lower bound that is approached only by impulses or unstable limits is an infimum, not an attained minimum in a class of ordinary stable responses.

For the quadratic calculation, assume the hypotheses of Proposition 5.11: in particular $e, v \in L^2$, $e(0) = 1$, and $e(t) \to 0$. These hold for the stable positive PI responses. For $\mathcal{E} = \int e^2$ and $\mathcal{U} = \int v^2$, the chain rule and $e' = -e - v$ give

$$\int ev = \frac{1}{2} - \mathcal{E}.$$

Cauchy–Schwarz therefore gives

$$\mathcal{U} \geq \phi(\mathcal{E}) := \mathcal{E} + \frac{1}{4\mathcal{E}} - 1, \qquad \mathcal{E} > 0. \tag{6.4}$$

The manuscript checks equality in the positive PI family by the exact identity

$$4\mathcal{E}(\mathcal{U}+1-\mathcal{E})-1=\frac{x(r-1)^2}{r(1+x)^2}.$$

Thus an off-line controller $r\neq 1$ is dominated by a controller on $r=1$ with the same tracking energy. On that line, $\mathcal{E}=1/(2x)$ and $\mathcal{U}=(x-1)^2/(2x)$. The portion $x<1$ is dominated by $(x,r)=(1,1)$; only $x\geq 1$ forms the efficient front. Indeed ϕ decreases strictly on $(0,1/2]$, so decreasing tracking energy there forces an increase in effort. This is why proving an equality curve is only part of proving a complete Pareto front.

6.3 The earliest admissible pulse minimizes an increasing moment

The command cone in manuscript Theorem 20 is described by $v\geq 0$ and $J_v=\int_0^\infty e^t v(t)\,\mathrm{d}t\leq 1$. A pointwise command cap imposes $v\leq c$, where $c>0$. The following argument derives its sharp consequence without any general rearrangement theorem.

Proposition 6.3 (A sharp moment bound under an amplitude cap)**.** *Let $0\leq v\leq c$ almost everywhere and $\int v=U<\infty$. Then, allowing an infinite left side,*

$$\int_0^\infty e^t v(t)\,\mathrm{d}t\geq c(e^{U/c}-1). \tag{6.5}$$

Equality holds exactly when $v=c\mathbf{1}_{[0,U/c)}$ almost everywhere.

Proof. The case $U=0$ follows from $v\geq 0$. For $U>0$, put $T=U/c$ and $v_*=c\mathbf{1}_{[0,T)}$. If the moment is infinite the inequality is immediate. Otherwise $\int(v-v_*)=0$, so

$$\int e^t(v-v_*)=\int(e^t-e^T)(v-v_*).$$

On $[0,T)$ both factors on the right are nonpositive. On (T,∞) both are nonnegative. The integral is therefore nonnegative, and the moment of v_* is $c(e^T-1)$. Equality forces $v=c$ almost everywhere before T and $v=0$ almost everywhere after T, because the weight is strictly increasing and its difference from e^T vanishes only at the singleton $\{T\}$. □

It follows that the feasible command area in the cone is exactly

$$0\leq U\leq U_c:=c\log(1+1/c)<1.$$

Every value in this interval is attained by the corresponding earliest pulse. On the efficient area line $A_e=S-U$, the squared MIACE is $(S-U)^2+\alpha U^2$, with derivative $2[(1+\alpha)U-S]$. The constrained optimum is therefore

$$U_{\mathrm{opt}}=\min\left\{\frac{S}{1+\alpha},U_c\right\}.$$

For $c=0$, only $v=0$ is possible. The bound is a statement about nonnegative centered commands in this cone; it is not a replacement for the separate classification of all amplitude-constrained PI laws.

6.4 Signs of two exponentials and repeated poles

Positivity of a sum of decaying exponentials is not equivalent to positivity of every coefficient. This distinction is central to the complete PI domain in manuscript Theorem 29.

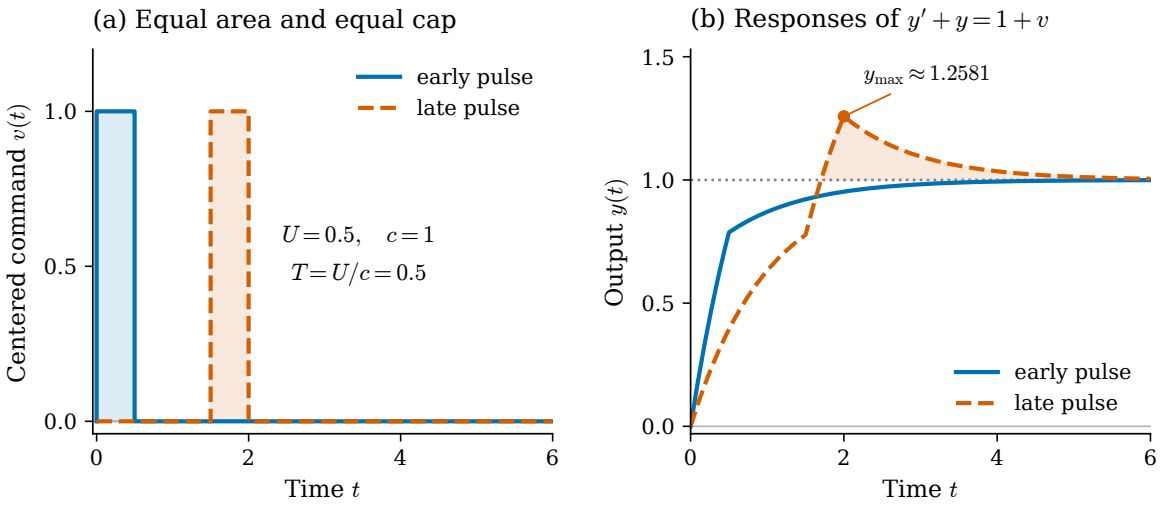


Figure 6.1: The same command area can have a different consequence when it is supplied later. Both pulses have height one and area 1/2. For the first-order plant, the early pulse has exponential moment $e^{1/2}-1<1$ and gives a monotone output. Delaying it by 1.5 multiplies that moment by $e^{1.5}$ and produces overshoot.

For positive PI parameters x, r, put $\Delta=(1+x)^2-4x/r$. If $\Delta>0$, the decay rates satisfy $0<\lambda_1<\lambda_2$ and

$$\lambda_{1,2}=\frac{1+x\mp\sqrt{\Delta}}{2}, \qquad e(t)=a_1e^{-\lambda_1 t}+a_2e^{-\lambda_2 t},$$

where

$$a_1=\frac{1-\lambda_1}{\lambda_2-\lambda_1}, \qquad a_2=\frac{\lambda_2-1}{\lambda_2-\lambda_1}.$$

Here $a_1+a_2=1$ and $\lambda_1a_1+\lambda_2a_2=x$. If $a_1<0$, the slower term eventually dominates and e becomes negative. If $a_1\geq 0$ and $a_2\geq 0$, both $e\geq 0$ and $-e'\geq 0$ are immediate. If $a_1\geq 0$ and $a_2<0$, factor out $e^{-\lambda_1 t}$. The remaining brackets in e and $-e'$ are

$$a_1+a_2e^{-(\lambda_2-\lambda_1)t}, \qquad \lambda_1a_1+\lambda_2a_2e^{-(\lambda_2-\lambda_1)t}.$$

Both increase; their initial values are 1 and $x>0$. They stay positive. Thus distinct real poles give nonovershoot exactly when $\lambda_1\leq 1$, and in that case the output is also monotone.

At $\Delta=0$, with $\lambda=(1+x)/2$, the repeated-pole formulas are

$$e=e^{-\lambda t}[1+(1-\lambda)t], \qquad -e'=e^{-\lambda t}[x+\lambda(1-\lambda)t].$$

They are nonnegative for all $t\geq 0$ exactly when $\lambda\leq 1$. If $\Delta<0$, the error is a nonzero exponentially damped sinusoid; it necessarily changes sign. Combining the cases gives

$$r\geq m(x), \qquad m(x)=\begin{cases}4x/(1+x)^2, & 0<x\leq 1,\\ 1, & x\geq 1.\end{cases} \tag{6.6}$$

For $x\leq 1$, real poles already imply $\lambda_1\leq 1$. For $x>1$, the inequality $\sqrt{\Delta}\geq x-1$ is equivalent to $r\geq 1$.

The same factoring method locates command extrema: after removing a positive exponential, a derivative is a constant plus one exponential. Its monotonic bracket has at most one zero. The coefficient signs must then be checked: for $r>1$ an interior command extremum is a minimum, whereas the monotone branch with $r<1$ has $v<0$. Together with the endpoint values, these

observations give the command peak bound in Theorem 29. For complex poles, the bracket is instead a sine–cosine combination; successive extrema have geometrically decreasing magnitudes. These are the elementary tools behind the manuscript's Appendix C.

Theorem 30 additionally proves that the whole monotone branch $x < 1$ is dominated, before minimizing over $1 \le x \le A$, $r \ge 1$. That comparison is needed to justify a global constrained optimum. Optimizing only the cancellation family would not establish it.

6.5 Taylor expansion of a squared residual

Suppose a smooth residual $R(\varepsilon)$ vanishes at the origin, $R(\varepsilon) = \ell(\varepsilon) + O(\|\varepsilon\|^2)$, where ℓ is linear. If a smooth denominator satisfies $D(\varepsilon) = D_0 + O(\|\varepsilon\|)$ with $D_0 > 0$, then

$$\frac{R(\varepsilon)^2}{D(\varepsilon)} = \frac{\ell(\varepsilon)^2}{D_0} + O(\|\varepsilon\|^3). \tag{6.7}$$

Indeed the cross term between ℓ and the quadratic remainder has order three, and $1/D = 1/D_0 + O(\|\varepsilon\|)$. Multiplying the latter correction by a quadratic numerator also gives order three. This explains why only first derivatives of the residuals are needed for Corollary 31.

Let $s = \sqrt{1+q_0} > 1$, $\kappa = 1 + \epsilon_k$, $\eta = 1 + \epsilon_\tau$, $x = \kappa s$, $r = 1/\eta$, and $\sigma = \sqrt{1 + q_0\kappa^2}$. Then

$$\sigma r - x = -\epsilon_k/s - s\epsilon_\tau + O(\|\epsilon\|^2), \qquad x - \sigma = \epsilon_k/s + O(\|\epsilon\|^2).$$

Applying (6.7) to the manuscript's two-square loss identity gives

$$L = \frac{\epsilon_k^2}{2s^3} + \frac{\epsilon_k\epsilon_\tau}{s(1+s)} + \frac{s\epsilon_\tau^2}{2(1+s)} + O((|\epsilon_k| + |\epsilon_\tau|)^3).$$

The coefficient matrix has positive diagonal and determinant $1/[4s(1+s)^2]$, so the leading quadratic form is positive definite.

Example 6.4 (A small cost loss with loss of nonovershoot)**.** With exact gain, $\kappa = 1$, the exact loss simplifies to

$$L(1, \eta) = \frac{s(\eta-1)^2}{2\eta(1+s)}.$$

Take $s = 3$ and $\eta = 1.05$. Then $L = 1/1120$, only $1/2240$ of the optimal normalized cost $s - 1 = 2$. Nevertheless $x = 3$ and $r = 1/1.05 < 1$, which violates (6.6). Stability persists, but nonovershoot does not. Continuity of an integral cost does not give a margin for a pointwise sign condition at a limiting trajectory whose error tends to zero.

6.6 Exercises with solutions

Exercise 6.5 (A negative coefficient that does not cause overshoot)**.** Can $\lambda_1 = 1/4$ and $\lambda_2 = 1/2$ be the two decay rates of a positive PI response? If not, find an admissible pair of distinct rates below one and prove its output monotone despite a negative error coefficient.

Solution. The rate sum gives $x = \lambda_1 + \lambda_2 - 1 = -1/4$, which is not positive. Thus these two rates cannot belong to the declared positive PI family. This check matters: arbitrary positive decay rates do not automatically define admissible PI parameters.

For an admissible instance, take instead $\lambda_1 = 3/5$ and $\lambda_2 = 4/5$. Then $x = 2/5$, $r = x/(\lambda_1\lambda_2) = 5/6$, $a_1 = 2$, and $a_2 = -1$. The factorizations

$$e = e^{-3t/5}(2 - e^{-t/5}), \qquad -e' = e^{-3t/5}(6/5 - 4e^{-t/5}/5)$$

are positive for all $t \ge 0$. The negative fast coefficient is compatible with a monotone response. □

Exercise 6.6 (A sharp cap and an unattainable area request)**.** For $S = 3/2$, $\alpha = 2$, and $c = 1/2$, determine whether the unconstrained equality areas can be attained within the nonnegative command cone under the cap. Then repeat for $\alpha = 1$.

Solution. The maximum area is $U_c = \frac{1}{2}\log 3 \simeq 0.5493$. At $\alpha = 2$, the desired area is $S/(1+\alpha) = 1/2 < U_c$. The pulse $v = \frac{1}{2}\mathbf{1}_{[0,1)}$ has this area and moment $(e-1)/2 < 1$, so it attains the equality areas. Its stable feedback realization is the internal-model construction of the manuscript. At $\alpha = 1$, the requested area is $3/4 > U_c$. The optimum under the cap instead uses $U = U_c$, with a pulse of duration $\log 3$. It is a constrained optimum above the unconstrained MIACE bound. □

Chapter 7

Nonlinear storage balances and exact MIACE tuning

Manuscript Theorem 32 keeps the same two absolute areas while allowing the plant's speed to depend on its state. Its proof needs no Laplace transform. A differentiable storage coordinate supplies the balance; the area decomposition from Chapter 6 then applies unchanged. The remaining work is to prove that the proposed feedback has a global solution, remains in the physical tracking interval, and produces integrable signals. This chapter supplies those steps.

Use the manuscript's fixed dimensionless time θ. The plant, error, and centered command are

$$y' = a(y)(w - y), \qquad y(0) = 0, \qquad e = 1 - y, \qquad v = w - 1, \tag{7.1}$$

where $a : \mathbb{R} \to (0, \infty)$ is locally Lipschitz. The target is one and there is no transport delay. The criterion is still

$$A_e = \int_0^\infty |e| \, \mathrm{d}\theta, \qquad A_u = \int_0^\infty |v| \, \mathrm{d}\theta, \qquad \mathcal{M}_\alpha = (A_e^2 + \alpha A_u^2)^{1/2}, \quad \alpha > 0.$$

For an ordinary locally integrable command, an admissible response is a globally defined, locally absolutely continuous y satisfying (7.1) almost everywhere, with $y(\theta) \to 1$ and $e, v \in L^1$. The existence argument below is needed for the particular attaining feedback; it does not silently assert existence for every arbitrary command.

7.1 A chain rule that can be integrated

The derivative calculation for a storage coordinate must remain valid when a command jumps. Absolute continuity is the relevant regularity.

Proposition 7.1 (Composition with a continuously differentiable function)**.** *Let $z : [b, c] \to \mathbb{R}$ be absolutely continuous, and let Φ be continuously differentiable on an open interval containing the range of z. Then $\Phi \circ z$ is absolutely continuous and*

$$\Phi(z(t)) - \Phi(z(b)) = \int_b^t \Phi'(z(s)) z'(s) \, \mathrm{d}s, \qquad b \le t \le c. \tag{7.2}$$

Proof. The range of the continuous function z is a compact interval J. Continuity of Φ' gives a finite bound $M = \max_J |\Phi'|$. The mean value theorem gives $|\Phi(z(t)) - \Phi(z(s))| \le M |z(t) - z(s)|$. Apply this estimate to any finite disjoint collection of intervals. The defining condition for absolute continuity of z then gives absolute continuity of the composition.

At every point where z' exists, the ordinary chain rule gives $(\Phi \circ z)' = \Phi'(z) z'$. Such points have full measure. Moreover $|\Phi'(z) z'| \le M |z'|$ is integrable. The fundamental theorem for absolutely continuous functions (Theorem 2.11) now gives (7.2). □

The compact interval in this proof is only the range on a finite time interval. No global bound on Φ' is needed. This matters because $1/a(y)$ need not be bounded on the whole real line.

7.2 Local solutions and why a bounded trajectory continues

A function F is locally Lipschitz when every point has a neighborhood on which $|F(x) - F(z)| \le L|x - z|$ for some finite L. This supplies uniqueness as well as existence. Continuity alone does not supply uniqueness; for example, a solution of $x' = \sqrt{|x|}$ can wait at zero before starting to increase.

Proposition 7.2 (Scalar existence, uniqueness, and compact continuation)**.** *If $F : \mathbb{R} \to \mathbb{R}$ is locally Lipschitz, the initial-value problem $x' = F(x)$, $x(0) = x_0$, has a unique local solution. Its maximal forward solution cannot have a finite terminal time while remaining in a fixed compact subset of $\mathbb{R}$.*

Proof. Choose $r > 0$ so that F is Lipschitz with constant L on $J = [x_0 - r, x_0 + r]$, and put $B = \max_J |F|$. Take $h > 0$ with $hB \le r$ and $q = hL < 1$. On continuous functions with $\|x - x_0\|_\infty \le r$, define

$$(\mathcal{T}x)(t) = x_0 + \int_0^t F(x(s))\,\mathrm{d}s, \qquad 0 \le t \le h.$$

The bound $hB \le r$ keeps this map in the same closed set, and $\|\mathcal{T}x - \mathcal{T}z\|_\infty \le q\,\|x - z\|_\infty$. Starting with the constant function x_0, successive iterates have differences bounded by a geometric sequence. They are uniformly Cauchy and converge uniformly to a continuous function in the same closed set. Passing to the limit under the integral gives a fixed point. The contraction inequality shows that two fixed points must coincide. A fixed point is continuously differentiable and solves the ODE. Repeating this argument and using uniqueness joins the local solutions into a maximal one.

Suppose its terminal time T_* is finite and $x(t)$ remains in a compact set K. The number $B_K = \max_K |F|$ is finite, so

$$|x(t) - x(s)| \le B_K|t - s|, \qquad 0 \le s, t < T_*.$$

Thus $x(t)$ has a limit x_* as $t \uparrow T_*$. Apply the local construction at x_*. Its new solution extends the old integral equation through T_*, contradicting maximality. □

This is the contraction proof of the scalar Picard–Lindelof theorem. Completeness here is elementary: a uniformly Cauchy sequence of continuous functions has a pointwise real limit, uniform convergence to that limit, and a continuous limit. It plays the same role as L^1 completeness in the convolution proof of Chapter 3.

For the attaining law below, $F(y) = c_\alpha a(y)(1 - y)$ is locally Lipschitz. Once its trajectory is shown to stay in $[0, 1]$, compact continuation gives existence for every nonnegative time. Positivity and continuity of a also give

$$0 < a_{\min} := \min_{[0,1]} a \le a(y) \le \max_{[0,1]} a =: a_{\max} < \infty. \tag{7.3}$$

These are bounds on the tracking interval. They do not assert a positive lower bound for a over all of $\mathbb{R}$.

7.3 The nonlinear balance and the unchanged area geometry

Proposition 7.3 (Storage balance for the original error and command)**.** *For the plant (7.1), define*

$$\Phi_a(y) = \int_0^y \frac{ds}{a(s)}, \qquad S_a = \Phi_a(1) = \int_0^1 \frac{ds}{a(s)}.$$

Every admissible response satisfies

$$\int_0^\infty e\,\mathrm{d}\theta + \int_0^\infty v\,\mathrm{d}\theta = S_a. \tag{7.4}$$

Writing $N = \int e_- + \int v_-$ *and* $m_\alpha^2 = \alpha/(1+\alpha)$*, it follows that*

$$\mathcal{M}_\alpha^2 - S_a^2 m_\alpha^2 = 4m_\alpha^2 N(S_a + N) + \frac{(A_e - \alpha A_u)^2}{1+\alpha}. \tag{7.5}$$

Proof. The function $1/a$ is continuous, so Φ_a is C^1. By Proposition 7.1, on every bounded time interval,

$$\frac{d}{d\theta}\Phi_a(y) = \frac{y'}{a(y)} = w - y = e + v \quad \text{almost everywhere.}$$

Integrate to time T, then let $T \to \infty$. The left side tends to $\Phi_a(1) - \Phi_a(0) = S_a$ by settling and continuity; the integral tends to the two signed areas by absolute integrability. The positive-part identity gives $A_e + A_u = S_a + 2N$. Substituting this into

$$A_e^2 + \alpha A_u^2 = \frac{\alpha(A_e + A_u)^2 + (A_e - \alpha A_u)^2}{1+\alpha}$$

proves (7.5), exactly as in Theorem 6.1. □

The storage coordinate has changed the way the differential equation is integrated. It has not changed the measured error to $\Phi_a(1) - \Phi_a(y)$, and it has not changed time. Both costs are still the integrals of $|1-y|$ and $|w-1|$ in the original dimensionless time θ.

Since $S_a > 0$, equality in (7.5) holds precisely when $N = 0$ and $A_e = \alpha A_u$. Thus $e, v \geq 0$ almost everywhere. The plant then gives $y' = a(y)(e+v) \geq 0$ almost everywhere; absolute continuity makes y nondecreasing at every time. Every response attaining this lower bound is therefore monotone. As before, area equality need not determine a unique trajectory or controller.

7.4 Constructing an attaining feedback

Proposition 7.4 (Invariant interval, integrable signals, and attainment)**.** *For* $\alpha > 0$*, put* $c_\alpha = 1 + 1/\alpha$ *and apply*

$$w = 1 + \frac{1-y}{\alpha}, \qquad v = \frac{e}{\alpha}. \tag{7.6}$$

The resulting solution exists for every $\theta \geq 0$*, stays in* $0 \leq y < 1$*, and satisfies*

$$e^{-c_\alpha a_{\max}\theta} \leq e(\theta) \leq e^{-c_\alpha a_{\min}\theta}. \tag{7.7}$$

It attains the global bound with

$$A_e = \frac{\alpha S_a}{1+\alpha}, \qquad A_u = \frac{S_a}{1+\alpha}, \qquad \mathcal{M}_\alpha = S_a\sqrt{\frac{\alpha}{1+\alpha}}. \tag{7.8}$$

Proof. The closed-loop equations are $y' = c_\alpha a(y)(1-y)$ and $e' = -c_\alpha a(y)e$. Along any local solution the latter gives

$$e(\theta) = \exp\left[-c_\alpha \int_0^\theta a(y(s))\,\mathrm{d}s\right].$$

It is positive at every finite time in the solution interval and no larger than one. Hence $0 \leq y < 1$ and $y' > 0$ there. Proposition 7.2 extends this bounded trajectory globally. Insert (7.3) in the exponential to obtain (7.7). In particular $e \to 0$, $e \in L^1$, and $v = e/\alpha \in L^1$.

The areas are nonnegative and $A_u = A_e/\alpha$. Together with (7.4) this gives (7.8). Every admissible response obeys the lower bound in (7.5), so attaining it proves global optimality over the declared class. □

The command starts at $1 + 1/\alpha$ and decreases to one. It has a finite jump from its zero prehistory and contains no impulse. The term one in (7.6) is equilibrium feedforward; the law is not asserted to be a positive PI controller driven only by error. It does not need the formula for a, but the assertion concerns exactly (7.1). The theorem does not cover an incorrect equilibrium command, extra states, or an added delay.

7.5 A nonlinear example with an exact implicit trajectory

For $a(y) = 1 + y^2$, the storage coordinate is $\Phi_a(y) = \arctan y$, so $S_a = \pi/4$. At $\alpha = 1$ the law is $w = 2 - y$ and

$$y' = 2(1 + y^2)(1 - y), \qquad A_e = A_u = \frac{\pi}{8}, \qquad \mathcal{M}_1 = \frac{\pi}{4\sqrt{2}}.$$

The areas are exact consequences of the balance, independently of any numerical integration of the trajectory.

For general α, separate the closed-loop equation to obtain

$$c_\alpha \theta = \int_0^{y(\theta)} \frac{dz}{(1 + z^2)(1 - z)} = -\frac{1}{2}\log(1 - y) + \frac{1}{4}\log(1 + y^2) + \frac{1}{2}\arctan y. \tag{7.9}$$

The derivative of the last expression is $1/[(1 + y^2)(1 - y)]$, its value at zero is zero, and it increases to infinity as $y \uparrow 1$. It therefore defines a unique trajectory for every $\theta \geq 0$.

For $a \equiv 1$, the same feedback gives $e = e^{-c_\alpha \theta}$ and $S_a = 1$. At $\alpha = 1$ its two areas are $1/2$ each. In both cases the area point lies on $A_e = A_u$, but its supporting line is $A_e + A_u = S_a$. The change from one to $\pi/4$ measures the effect of the nonlinear storage coefficient, with no alteration of the criterion. Manuscript Figure 13 compares these trajectories and records the associated optimal areas.

7.6 Finite-horizon integration and its omitted tail

Numerical integration stops at a finite time, whereas MIACE uses an infinite horizon. For the attaining law, define $R_e(T) = \int_T^\infty e(\theta)\,\mathrm{d}\theta$ and $R_u(T) = \int_T^\infty v(\theta)\,\mathrm{d}\theta$. Since $e + v = c_\alpha e$, integrating the storage identity from T to infinity gives the exact formula

$$R_e(T) = \frac{\Phi_a(1) - \Phi_a(y(T))}{c_\alpha}, \qquad R_u(T) = \frac{R_e(T)}{\alpha}. \tag{7.10}$$

Using (7.3) in the spatial integral gives

$$\frac{e(T)}{c_\alpha a_{\max}} \leq R_e(T) \leq \frac{e(T)}{c_\alpha a_{\min}}. \tag{7.11}$$

The same bounds follow by integrating $e(T)e^{-c_\alpha a_{\max} s} \leq e(T + s) \leq e(T)e^{-c_\alpha a_{\min} s}$ over $s \geq 0$.

Write $A_e(T) = \int_0^T e$ and $A_u(T) = \int_0^T v$, and let $\mathcal{M}_\alpha(T)$ use these truncated areas. Their ratio is also $A_u(T) = A_e(T)/\alpha$, so

$$\mathcal{M}_\alpha - \mathcal{M}_\alpha(T) = \sqrt{c_\alpha}\, R_e(T) \leq \frac{e(T)}{\sqrt{c_\alpha}\, a_{\min}}. \tag{7.12}$$

These formulas bound the analytical omitted tail. Errors in an approximate value of $y(T)$ or in the numerical quadrature require separate bounds.

7.7 Exercises with solutions

Exercise 7.5 (Constant storage and the time scale)**.** Take $a(y) = a_0 > 0$. Find the trajectory, both optimal areas, and the minimum MIACE. Explain why changing a_0 does not change the attaining feedback gain.

Solution. Here $S_a = 1/a_0$ and $e(\theta) = e^{-c_\alpha a_0 \theta}$, $v = e/\alpha$. Thus $A_e = \alpha/[a_0(1+\alpha)]$, $A_u = 1/[a_0(1+\alpha)]$, and $\mathcal{M}_\alpha = a_0^{-1}\sqrt{\alpha/(1+\alpha)}$. The rate a_0 scales both areas together; the optimal ratio remains $A_e/A_u = \alpha$. The feedback is still $w = 1 + (1-y)/\alpha$. □

Exercise 7.6 (An explicit stopping bound)**.** For $a(y) = 1 + y^2$ and $\alpha = 1$, suppose the exact output at a stopping time is $y(T) = 0.999$. Bound the missing tracking area and MIACE, and give an exact expression for the missing area.

Solution. On $[0, 1]$, $a_{\min} = 1$, $a_{\max} = 2$, and $c_\alpha = 2$. Therefore

$$0.00025 \le R_e(T) \le 0.0005, \qquad \mathcal{M}_1 - \mathcal{M}_1(T) \le \frac{0.001}{\sqrt{2}}.$$

The exact missing area is $R_e(T) = [\pi/4 - \arctan(0.999)]/2$, and the missing command area is identical. The numerical bounds use only the endpoint error and the two compact bounds on a. □

Exercise 7.7 (Why positivity at the target matters)**.** Drop strict positivity at $y = 1$ and take $a(y) = |1 - y|$. Apply the same feedback from $y(0) = 0$. Does the output still converge monotonically? Are its MIACE areas finite?

Solution. While $0 \le y < 1$, $a(y) = 1 - y = e$. Hence $e' = -c_\alpha e^2$ and $e(\theta) = 1/(1 + c_\alpha\theta)$. The output does converge monotonically to one, but $\int_0^\infty e = \infty$, and $v = e/\alpha$ also has infinite area. Correspondingly $\int_0^1 a(y)^{-1}\,\mathrm{d}y$ diverges. This plant violates the strict positivity hypothesis of manuscript Theorem 32. Settling and monotonicity alone do not replace the integrability proof. □

Chapter 8

A reading map for every numbered result

The first column below uses the manuscript's numbering and exact result names. The middle column points to background developed in this companion; the last identifies the step to follow in the manuscript proof. Thus a reference such as corollary 3.10 is a companion reference, whereas "result 9" always means manuscript Lemma 9. Section numbers prefixed by "paper" also refer to the manuscript. The map supplies prerequisites and explains dependencies; the manuscript remains the source of the complete tuning statements, parameter restrictions, and constructions.

8.1 The 32 results

Manuscript result	Companion prerequisites	Key step in the manuscript proof
1. Theorem. Complete L^1 front and normalized MIACE optimum	Integral equality: proposition 2.2; quadratic identity: theorem 6.1; pole signs: section 6.4.	In paper §3, bound absolute areas by signed areas; minimize along that boundary. Appendix B excludes every noncancelling PI controller from equality at an interior point.
2. Theorem. Exact MIACE excess and shape of every minimizer	Absolute continuity: theorem 2.11; negative parts: proposition 2.2; balance and square completion: theorem 6.1 and section 6.1.	Integrate the plant equation, then separate the exact excess into sign-reversal and area-ratio terms. Equality forces $e, v \geq 0$, hence $y' = e + v \geq 0$.
3. Theorem. MIACE controls output variation and overshoot	Total variation and endpoint recovery: proposition 2.13; weighted area inequality: theorem 6.1.	Bound $\int \lvert y' \rvert$ by $A_e + A_u$. An excursion above the final value costs both the upward excursion and the return in total variation.
4. Proposition. Finite-record overshoot bound	Finite-interval fundamental theorem and variation: theorem 2.11 and proposition 2.13.	Retain the measured terminal error. The path from the initial value to its maximum and then to the terminal value bounds the recorded peak.
5. Theorem. MIACE excess and shape with pure delay	Causal shifts: proposition 4.3; signed-area balance and equality: section 6.1 and theorem 6.1.	Separate the waiting interval, whose error area is δ. Apply the same nonnegative-part identity to the shifted plant motion; the signed-area sum becomes $1 + \delta$.
6. Lemma. Stability and monotonicity on $r = 1$	Method of steps: equation (5.4); stability and positivity: theorem 5.8, proposition 5.9, and equation (5.5).	Reduce to $g' = -xg(\cdot - \delta)$ with zero negative prehistory and $g(0) = 1$. Stability has threshold $x\delta < \pi/2$; nonnegativity has threshold $x\delta \leq 1/e$.
7. Theorem. Exact delayed PI optimum and its attainment range	Analytic transforms and positivity: theorems 4.5 and 4.8, proposition 4.7, and example 4.9; result 6.	Equality gives an exponential bound on the shifted error. The transform identity then forces $r = 1$; signs, the delay threshold, and the optimal area ratio fix the remaining range.

Manuscript result	Companion prerequisites	Key step in the manuscript proof
8. Proposition. Structure of an attaining genuine PID controller	Real-axis transform positivity and analytic continuation: proposition 4.7 and theorem 4.8; retained modes: section 5.3.	Use the filter zero and the value at the plant pole to exclude forbidden filter parameters and force the necessary cancellation and gain relations.
9. Lemma. Nonnegative convolution resolvent	Completeness: theorem 3.2; Young's inequality and positive Neumann series: theorems 3.4 and 3.9 and corollary 3.10.	Sum $f + K * f + K^{*2} * f + \cdots$. The kernel mass below one gives convergence and uniqueness; every term preserves nonnegativity.
10. Theorem. A two-rate sufficient condition for the delayed PID optimum	Positive resolvents and exponential weights: corollary 3.10, lemma 3.11, and proposition 3.12; theorem 5.7.	Construct a nonnegative finite-support kernel of mass below one and nonnegative command forcing. Factor the full characteristic expression to verify internal exponential stability.
11. Theorem. An explicit additional attainment interval	Result 10; elementary exponential estimates and positivity of integrals from chapter 2.	Substitute the proposed two-rate family into every sufficient inequality of result 10. The displayed algebraic bounds give an explicit additional delay interval.
12. Theorem. MIACE excess for a general stable plant	Transform value at zero: proposition 4.2; dominated convergence: theorem 2.6; balance identity: section 6.1 and theorem 6.1.	Extract the signed-area identity from the low-frequency plant expansion. Apply the exact area decomposition with $S = \mu + \delta$; positivity of a cascade is not needed for this identity.
13. Lemma. One-way crossing through a cascade	Variation of constants and positivity of lags: propositions 5.1 and 5.2 and lemma 5.3.	Once a stage exceeds one, a command at least one prevents it from crossing back below one. Propagate the resulting excursion through the downstream positive stages.
14. Theorem. Shape of every attaining cascade response	Result 13; derivative signs for absolutely continuous functions: corollary 2.12; lag monotonicity: proposition 5.2.	Equality gives nonnegative error and centered command. One-way crossing excludes an intermediate overshoot, and the lag equations then yield monotonicity through the cascade.
15. Theorem. Necessary condition for attaining the bound	Fatou's lemma: lemma 2.5; strict moment inequality: section 2.7; positive convolution: equation (3.2).	Take the positive-kernel tail ratio to obtain $\int e^t v \le 1$. For a nonzero ordinary command this implies $\int v < 1$, yielding the strict weight–delay condition.
16. Lemma. Exact delay remainder	Hermite interpolation, simplex kernels, and causal shifting: propositions 4.10 and 4.11, theorem 4.12, and equation (4.3).	Match every pole's principal part, including multiplicities. The residual inverse transform vanishes beyond the delay, giving an exact finite-support remainder with a controlled sign.
17. Theorem. Sufficient conditions for global attainment	Result 16; corollary 3.10 and lemma 3.11; internal stability: theorem 5.7 and section 5.3.	Use the exact remainder in a denominator factorization. Its kernel and the command forcing satisfy positivity and small-mass bounds; check the full retained dynamics before invoking the area equality.
18. Corollary. An attaining order-N controller for small delays	Result 17; continuity of finite-support integrals: theorems 2.6 and 4.12; proper realizations: section 5.3.	Strict inequalities at zero delay persist for small positive delay. The nonzero numerator value at the candidate cancellation point ensures the asserted controller order.

Manuscript result	Companion prerequisites	Key step in the manuscript proof
19. Theorem. Uniform construction for every $N \geq 2$	Repeated poles and simplex formulas: equation (4.1) and proposition 4.11; moment integration: example 2.9; result 17.	Equal lag times give an explicit exponential-polynomial remainder. Bound its moments uniformly in N, verify all construction inequalities, and exclude an order-reducing cancellation.
20. Theorem. The nonovershooting command cone	Exponential moments: section 2.7; Fatou: lemma 2.5; positive lag propagation: proposition 5.2.	Necessity uses the tail-ratio argument. For sufficiency, put the slowest lag first: its explicit solution stays below one exactly under the moment bound; positivity handles the remaining lags.
21. Theorem. Exact attainment when internal delays are allowed	Result 20; stable-block internal-model realization: section 5.5; area equality: section 6.1 and theorem 6.1.	Choose a stable shaping filter giving nonnegative error and command with the required area ratio. Retain the model states and histories when establishing internal stability.
22. Theorem. Sharp amplitude-cap criterion	Increasing-weight rearrangement and equality: section 6.3; command-cone moment condition: result 20.	Among commands with given area and height cap, the initial rectangular pulse minimizes the exponential moment. This gives the sharp capacity and its unique equality shape a.e.
23. Theorem. Delay-error stability and MIACE deviation	Finite-measure filters and inverse series: theorems 3.7 and 3.9; delay and norm estimates: example 3.13, equation (3.5), and proposition 3.14.	Estimate the smoothed delay difference, then invert the perturbed loop by a norm-convergent Neumann series. Apply signal-norm bounds to the two absolute areas and their weighted Euclidean norm.
24. Lemma. Second-order energy formula	Exponential matrix decay and Gramians: propositions 5.5 and 5.12 and equation (5.7).	Solve the 2×2 Lyapunov equation and evaluate $CPC^\top$. The calculation includes repeated poles without taking differences of singular residue formulas.
25. Proposition. Exact PI energies	Result 24; the difference between absolute and squared areas: chapter 2; realizations: section 5.3.	Insert the normalized error and centered-command numerators into the second-order formula, keeping the shared denominator and normalization consistent.
26. Theorem. Unique global quadratic optimum	L^2 inner products and exact scalar completion: section 5.6 and equation (5.6); result 25.	Complete squares in the signal cost and, equivalently, in the PI parameter formula. Vanishing of the nonnegative remainder gives the unique gains and the feedback relation.
27. Proposition. Delayed quadratic energies on the cancellation family	Scalar delay stability and autocorrelation: theorem 5.8, section 5.8, and equation (5.8).	Differentiate the decaying correlation under its integral, obtain its delay relation and boundary value, and recover both energies. Use only the stable cancellation family.
28. Theorem. Complete L^2 Pareto front	Supporting weighted objectives and dominance: section 6.2; results 25–26.	Eliminate the cancellation-family parameter to obtain the front. Strict decrease of the exact lower boundary establishes efficiency; comparisons exclude the remaining PI points from the efficient set.

Manuscript result	Companion prerequisites	Key step in the manuscript proof
29. Theorem. Complete nonovershooting domain	Distinct, repeated, and complex pole signs: section 6.4 and equation (4.1); absolute continuity: corollary 2.12.	Classify the sign of the error by the slowest residue or the repeated-pole polynomial. Complex poles force sign changes; monotonicity follows within this specific positive PI family.
30. Theorem. Amplitude-constrained optima without overshoot	Result 29; one-variable constrained minimization and frontier geometry: section 6.2; peak formulas in paper Appendix C.	Convert the command cap into a restriction on the nonovershooting PI domain. Monotonicity in one parameter reduces optimization to the boundary, then to clipping the scalar optimum.
31. Corollary. Mismatch loss and its local expansion	Exact quadratic optimum: result 26; multivariable Taylor expansion: section 6.5.	Substitute the gain and time-constant mismatch ratios into the exact loss. Expand the resulting smooth rational function at the nominal point, where the first-order terms vanish.
32. Theorem. Exact MIACE optimum for a nonlinear storage class	Absolutely continuous chain rule: proposition 7.1; existence and continuation: proposition 7.2; storage balance and attainment: propositions 7.3 and 7.4.	In paper §15, integrate $\Phi_a(y)' = e + v$ to obtain the intercept S_a. Apply the same exact area decomposition. Prove the output stays in $[0, 1]$, the error decays exponentially, and the equality conditions hold. The class has zero delay.

8.2 The appendices and the numerical section

Appendix A: the second-order energy formula. This supplies the proof used by result 24, rather than an additional numbered theorem. The companion develops both the general integral Gramian argument and the entrywise 2×2 calculation in proposition 5.12 and equation (5.7). The essential prerequisite is exponential decay of the matrix exponential, which makes the improper integral converge and removes its boundary term at infinity.

Appendix B: equality in the two absolute-area bounds. This is the uniqueness argument supporting result 1. First turn equality of absolute and signed areas into pointwise nonnegativity, using continuity and proposition 2.2. Then separate $r > 1$, $r < 1$ with real poles, repeated poles, and complex poles. In each excluded case an error or command term must eventually become negative. The slowest-exponential and repeated-root arguments are explained in section 6.4 and equation (4.1).

Appendix C: extrema, absolute areas, and numerical checks. Differentiate a sum of two exponentials to find its possible stationary point; use the exponential times a linear polynomial at a repeated pole. For complex poles, successive extrema alternate and decrease geometrically. Absolute areas require splitting at zeros before integrating: the damped sinusoid's infinite tail is a geometric sum of lobe areas. These are applications of section 6.4 and proposition 2.2; the state-space energy check uses proposition 5.12. The formulas are exact mathematical identities. Their floating-point evaluation and ODE cross-checks are numerical evidence, not rigorous interval enclosures.

Section 14: numerical examples and verification. Read its first four subsections beside results 2–5, 7, 10–11 and 17–19: they illustrate shape bounds, delay thresholds, filtered PID attainment, and arbitrary-order constructions. The dimensional example requires restoring the time and plant-gain scaling from paper §2. The constrained search and independent checks use Appendix C and the energy formulas. A finite grid, an optimizer's output, or agreement of two floating-point implementations does not prove a global optimum or an infinite-horizon sign condition. Those conclusions come from the corresponding analytic results; the computations test

their implementation and illustrate their size.

Section 15: the nonlinear example and Figure 13. Theorem 32 uses the same measured error, centered command and time variable as the original MIACE objective. The storage coordinate integrates the plant equation; it does not redefine the error norm. The example $a(y) = 1 + y^2$ is worked through in section 7.5, including its implicit trajectory and exact areas. The companion's tail estimate in section 7.6 explains how a finite numerical record can be compared with those infinite-horizon formulas. As in Section 14, the plotted trajectories illustrate the proof rather than establish it.

8.3 What is proved here, and what remains standard background

The companion proves the area identities, the relevant integral estimates, convolution inverse construction, interpolation mechanism, positive-lag arguments, scalar square completion, and Gramian calculation. These explain the manuscript's proof mechanisms. Chapter 7 adds the absolutely continuous chain rule, scalar ODE continuation, nonlinear storage balance and explicit attainment argument. Reproducing these tools here is not a claim that their standard mathematical ingredients are new. The manuscript's specific tuning formulas and controller constructions retain their own hypotheses and proofs.

The following substantial background inputs are explicitly retained:

1. **Measure and real analysis.** The construction of Lebesgue measure and product measure is assumed. Tonelli–Fubini is stated with its sign or absolute-integrability hypotheses in theorem 2.8; its full measure-theoretic construction is not reproduced. The absolute continuity theorem in theorem 2.11 uses Lebesgue differentiation and the characterization of absolutely continuous functions [5, 12]. Monotone convergence, Fatou's lemma and dominated convergence are proved from the integration framework.

2. **Complex analysis and transform uniqueness.** Local Taylor expansions of holomorphic functions are standard background; they give the identity theorem proved in theorem 4.8. Laplace uniqueness in theorem 4.6 is reduced to Fourier uniqueness on L^1, which is cited rather than reproved [7, 10]. The positivity and weighted analyticity estimates needed by the paper are developed directly.

3. **Retarded differential equations.** The equivalence between exponential stability and a strictly negative characteristic spectral bound, for the finite-dimensional constant-coefficient retarded systems specified in theorem 5.7, is a classical theorem [8]. The full scalar characteristic-root classification behind $x\delta < \pi/2$ uses Hayes [9]; the elementary stable-side and crossing calculations are given in theorem 5.8. The converse oscillation criterion at $x\delta > 1/e$ is also a classical input, identified in equation (5.5); the continuous-delay statement is recalled in the introduction of Benekas et al. [2]. The sufficiency of $x\delta \leq 1/e$ for the specified fundamental solution is proved in proposition 5.9. These two thresholds answer different questions and must not be interchanged.

Standard finite-dimensional linear algebra, including Jordan form, and ordinary calculus are assumed throughout. The matrix decay argument, retained-state distinction, scalar optimal-control identity and Lyapunov calculation are supplied in Chapter 5; a general infinite-dimensional spectral theorem or a general optimal-control existence theorem is not required for these proofs.

References and further reading

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